

Math 40003: Linear Algebra and Groups

Term 1 Lecture Notes

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1 Introduction to vector spaces and fields

Motivation: Linear algebra is ubiquitous: its applications are never-ending, particularly to computers and signal processing (audio and video samples, data, and communications), as well as to 2D and 3D geometry.

Let F be a field, e.g., $\mathbb{R}, \mathbb{C}, \mathbb{Q}, \mathbb{F}_2, \dots$. We are interested in *vector spaces* V over F . We will give the definition shortly, but let us consider some important examples of fields and vector spaces over them.

In 2D geometry we often study:

$$F = \mathbb{R}, \quad V = \mathbb{R}^2 = \{x = (x_1, x_2) : x_1, x_2 \in \mathbb{R}\}.$$

In 3D geometry we often study:

$$F = \mathbb{R}, \quad V = \mathbb{R}^3 = \{x = (x_1, x_2, x_3) : x_1, x_2, x_3 \in \mathbb{R}\}.$$

In computer science we often study:

$$\begin{cases} \mathbb{F}_2^N, N \gg 0 \\ \text{(or } \mathbb{F}_p^N, \text{ more generally)} \end{cases}$$

Definition 1.1. The vector space F^n of n -tuples of elements of F is defined as:

$$F^n = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_1, \dots, x_n \in F \right\}, \quad F = \text{a field.}$$

Remark 1.2. Pedantically, we should define F^n using the Cartesian product construction:

$$F^n := \underbrace{F \times \cdots \times F}_{n \text{ times}} = \{n\text{-tuples } (x_1, \dots, x_n) \text{ of elements } x_i \in F\}$$

This is also not completely precise, since the Cartesian product is usually first defined for products of two sets (so, the case $n = 2$), before getting to products of n sets. So an even more pedantic definition could be:

$$F \times (F \times (\cdots \times (F \times F) \cdots)) = \{(x_1, (x_2, \dots, (x_{n-1}, x_n) \dots)) : x_i \in F\},$$

or any other way of bracketing the n copies of F into Cartesian products.

In this course, as in most of mathematics, such distinctions will be ignored, except in discussions of the foundations of mathematics, or in situations where bracketing is important (due to a lack of associativity of some structure). We will simply assume the well-known fact that F^n is well-defined and its elements are as described in Definition 1.1.

Example 1.3. In the case $n = 1$, we can make identify F^1 with F by removing brackets:

$$F^1 \xrightarrow{\sim} F, (a) \mapsto a.$$

In the first problem sheet (PS1), you will prove that $V = F^n$ is a vector space. The definition of vector space is as follows. We begin with the definition of a commutative group:

Definition 1.4. A commutative group $(V, +)$ is a set V equipped with a binary operation $+$: $V \times V \rightarrow V$ satisfying:

- For $u, v, w \in V$, $u + (v + w) = (u + v) + w$ (associativity)
- For $v, w \in V$ $v + w = w + v$ (commutativity)
- For $v \in V$ $\exists 0 := (0, \dots, 0) \in V$, $v + (0, \dots, 0) = (0, \dots, 0) + v$. (zero vector exists)
- For $v \in V$, $\exists -v \in V$ $v + (-v) = 0 = (-v) + v$ (inverses exist)

Remark 1.5. Note that the element 0 is unique, since given another zero $0'$, we have $0' = 0 + 0' = 0$. Also, additive inverses (negatives) are unique as well (provided they exist), because if $w + v = 0$, then $v + (-v) = 0 = w + v$, which implies that $w = w + 0 = w + (v + (-v)) = (w + v) + (-v) = 0 + (-v) = -v$.

Definition 1.6. A vector space V over a field F is a commutative group $(V, +)$ together with an action $F \times V \rightarrow V$, $(\lambda, v) \mapsto \lambda v$, called scalar multiplication, satisfying the following properties: For all $\lambda, \mu \in F$ and $v, w \in V$,

- $\lambda(\mu v) = (\lambda\mu)v$ (associativity)
- $\lambda(v + w) = \lambda v + \lambda w$, $\lambda \in F, v, w \in F^n$ (distributivity)
- $(\lambda + \mu)v = \lambda v + \mu v$ (distributivity)
- $1 \cdot v = v$ (unit multiplication).

As an example, let us verify the unit multiplication property:

$$1 \cdot (v_1, \dots, v_n) = (1 \cdot v_1, \dots, 1 \cdot v_n) = (v_1, \dots, v_n).$$

Other examples of vector spaces:

- $\cdot F \subseteq E$ (fields $\Rightarrow E$ is an F -VS)
- $\cdot F[X] = \{a_0 + a_1X + \dots + a_nX^n : a_i \in F\}$ polynomials
- $\cdot \text{Fun}(X, F) = \{f : X \rightarrow F\}$ functions

Remark 1.7. These examples also have a commutative multiplication of vectors by vectors, but in general vector spaces we cannot multiply two vectors.

Definition 1.8. A subset $W \subseteq V$ of vector space V is called a *subspace* if:

- $0 \in W$
- W is closed under addition, i.e., $v, w \in W$ implies $v + w \in W$
- W is closed under scalar multiplication: for all $w \in W$ and $\lambda \in F$, we have $\lambda w \in W$.

Observation 1.9. $(-1) \cdot v = -v$: see PS1 for more on this exercise.

Therefore, W is also a vector space.

Observation 1.10. For every $w \in W$, we have $0 \cdot w = 0$, because $0 \cdot w = (0 + 0) \cdot w = 0 \cdot w + 0 \cdot w$, and subtracting $0 \cdot w$ from both sides yields $0 = 0 \cdot w$.

Using the observation, we can replace the axiom that $0 \in W$ by the axiom that $W \neq \emptyset$.

Recollections on the definition of fields and related notions

Definition 1.11. A *semigroup* is a set S together with a binary operation

$$S \times S \rightarrow S, \quad (a, b) \mapsto a \cdot b,$$

which is associative: for all $a, b, c \in S$ we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Definition 1.12. A *monoid* is a semigroup M together with a distinguished element $1 \in M$, called the *identity*, such that

$$a \cdot 1 = 1 \cdot a = a \quad \text{for all } a \in M.$$

Definition 1.13. A *group* is a monoid G such that for every element $g \in G$ there exists an element $g^{-1} \in G$, called the *inverse* of g , with

$$gg^{-1} = 1 = g^{-1}g.$$

Definition 1.14. A group G is called *commutative* (or *abelian*) if

$$gh = hg \quad \text{for all } g, h \in G.$$

Definition 1.15. A *ring* $(R, +, \cdot)$ is a set R equipped with two binary operations, addition $+$ and multiplication $\cdot$, such that:

1. $(R, +)$ is a commutative group. In particular, addition is associative and commutative, there is an additive identity element $0 \in R$, and every $a \in R$ has an additive inverse $-a$.
2. $(R, \cdot)$ is a monoid. In particular, multiplication is associative and there is a multiplicative identity element $1 \in R$.
3. Multiplication distributes over addition: for all $a, b, c \in R$ we have

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \text{and} \quad (a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Definition 1.16. A ring $(R, +, \cdot)$ is called *commutative* if the multiplication is commutative, that is,

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Definition 1.17. A *field* is a commutative ring R such that $(R \setminus \{0\}, \cdot)$ is a group. In other words, every non-zero element of R has a multiplicative inverse.

Definition 1.18. Let E be a field. A *subfield* of E is a subset $F \subseteq E$ such that:

1. $1 \in F$;
2. F is closed under addition and multiplication: if $a, b \in F$, then $a + b \in F$ and $ab \in F$;
3. F is closed under additive inverses: if $a \in F$, then $-a \in F$;
4. F is closed under multiplicative inverses of non-zero elements: if $a \in F$ and $a \neq 0$, then $a^{-1} \in F$.

Observation 1.19. In any field we may write sums and products without parentheses, because both operations are associative. Thus

$$a + b + c = (a + b) + c = a + (b + c) \quad \text{and} \quad abc = (ab)c = a(bc).$$

We also define subtraction in terms of addition by

$$a - b := a + (-b).$$

Observation 1.20. You may use without proof that $\mathbb{R}$ and $\mathbb{Q}$ are fields; this should be familiar from earlier courses.

More examples of fields

Definition 1.21. Let p be a prime number. The set

$$\mathbb{F}_p = \{0, 1, \dots, p-1\}$$

becomes a field when we define addition and multiplication *modulo* p . More precisely, to add or multiply two elements $a, b \in \mathbb{F}_p$, we first perform the usual integer addition or multiplication, and then take the remainder upon division by p , which is an element of $\{0, 1, \dots, p-1\}$. Subtraction is defined similarly by working in the integers and then reducing modulo p .

Exercise (Problem Sheet 1). Prove that $\mathbb{F}_p$ is a field.

Hints for the exercise. First check that $(\mathbb{F}_p, +, 0)$ is a commutative group: addition is commutative and associative, 0 is an additive identity, and every $a \in \mathbb{F}_p$ has an additive inverse (namely $p - a$, reduced modulo p). Next check that $(\mathbb{F}_p, \cdot, 1)$ is a commutative monoid: multiplication is associative and commutative, and 1 is a multiplicative identity. It is also straightforward to verify that multiplication distributes over addition:

$$a(b + c) = ab + ac \quad \text{and} \quad (a + b)c = ac + bc$$

for all $a, b, c \in \mathbb{F}_p$.

The less obvious step is to show that every non-zero element has a multiplicative inverse.

Claim 1.22. For every $a \in \mathbb{F}_p$ with $a \neq 0$, there exists $a^{-1} \in \mathbb{F}_p$ such that

$$aa^{-1} = 1 = a^{-1}a.$$

Proof. Because multiplication in $\mathbb{F}_p$ is commutative, it is enough to find $b \in \mathbb{F}_p$ such that $ab = 1$. Consider the p elements

$$a \cdot 0, a \cdot 1, a \cdot 2, \dots, a \cdot (p-1)$$

of $\mathbb{F}_p$. Each of these is well-defined modulo p . We claim that these p elements are all distinct. Suppose that

$$a \cdot c = a \cdot d \quad \text{for some } c, d \in \{0, 1, \dots, p-1\}.$$

Then

$$a \cdot (c - d) = 0 \quad \text{in } \mathbb{F}_p.$$

The following lemma shows that this implies $c - d \equiv 0 \pmod{p}$, hence $c = d$, so the list above has p distinct elements. Since $\mathbb{F}_p$ has exactly p elements, these p multiples of a must be all of $\mathbb{F}_p$. In particular, 1 is one of them, so there exists $b \in \mathbb{F}_p$ such that $ab = 1$, as required. $\square$

Lemma 1.23. Let $x, y \in \mathbb{F}_p$. Then

$$xy = 0 \iff x = 0 \text{ or } y = 0.$$

Proof. Think of x and y as integers in $\{0, 1, \dots, p-1\}$. The equality $xy = 0$ in $\mathbb{F}_p$ means that p divides the integer product xy . Since p is prime, this implies by a standard result you have seen that p divides x or p divides y . Thus $x \equiv 0 \pmod{p}$ or $y \equiv 0 \pmod{p}$, which means $x = 0$ or $y = 0$ in $\mathbb{F}_p$. $\square$

Using the claim and the lemma, we conclude that every non-zero element of $\mathbb{F}_p$ has a multiplicative inverse, and therefore $\mathbb{F}_p$ is a field.

We now describe some further examples of fields.

- The set

$$\mathbb{Q}[i] = \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$$

is a field, where i is a complex number with $i^2 = -1$ (as usual). Addition and multiplication are defined by the usual rules for complex numbers.

- We can construct a similar example starting from $\mathbb{F}_p$ instead of $\mathbb{Q}$. Suppose that the equation

$$x^2 + 1 = 0$$

has no solution in $\mathbb{F}_p$. This happens precisely when $p \equiv 3 \pmod{4}$, that is, when $p = 4k + 3$ for some integer k . In that situation we can adjoin a symbol α satisfying $\alpha^2 + 1 = 0$ and consider

$$\mathbb{F}_p[\alpha] = \{a + b\alpha : a, b \in \mathbb{F}_p\}.$$

One can check that this is a field with p^2 elements.

- In the field $\mathbb{F}_p[\alpha]$ we still perform addition and multiplication modulo p on the coefficients a and b . In particular, $p \cdot 1 = 0$ in this field, so we are *not* working modulo p^2 (which would give the ring $\mathbb{Z}/p^2\mathbb{Z}$). Instead, we have constructed a new field of size p^2 .

- Given any complex number $z \in \mathbb{C}$, we can consider the smallest subfield of $\mathbb{C}$ that contains both $\mathbb{Q}$ and z . This is called the *subfield generated by z* and is usually denoted by $\mathbb{Q}(z)$. It consists of all rational functions in z with coefficients in $\mathbb{Q}$:

$$\mathbb{Q}(z) = \left\{ \frac{p(z)}{q(z)} \mid p, q \in \mathbb{Q}[X], q(z) \neq 0 \right\},$$

where $\mathbb{Q}[X]$ denotes the ring of polynomials with rational coefficients.

- Analogously, starting from $\mathbb{F}_p$ one can construct finite fields of size p^m for any prime p and any positive integer m , by adjoining roots of suitable polynomials. These fields are often denoted by $\mathbb{F}_{p^m}$.
- A fundamental fact (which we will not prove here) is that for each prime power p^m , any two fields with p^m elements are isomorphic. In other words, up to isomorphism there is essentially only one field of size p^m .

In these lectures we will develop matrix theory and linear algebra over an *arbitrary* field. You can keep your favourite example (for instance $\mathbb{R}$, $\mathbb{Q}$, or $\mathbb{F}_p$) in mind while we work.

There are also important results that are specific to particular fields such as $\mathbb{R}$ or $\mathbb{C}$, for example spectral theorems or various decomposition results. We will not study those results in this course.

Practical information.

- You can get AI-generated feedback on your solutions to problems from the problem sheets at <https://www.lambdafeedback.com>.
- You can try to code solutions to the first problem sheet in the theorem prover Lean. A starting point is the repository <https://github.com/ImperialCollegeLondon/LAGinLean2024/tree/main/LAGinLean/>.

2 Matrices, linear equations, and Gaussian elimination

Matrices as coordinates for linear transformations

Matrices give a convenient “coordinate” description of linear transformations.

Let $T : V \rightarrow W$ be a linear map between vector spaces over a field F . Recall that linearity means

$$T(v_1 + v_2) = T(v_1) + T(v_2) \quad \text{and} \quad T(\lambda v) = \lambda T(v)$$

for all $v_1, v_2, v \in V$ and all scalars $\lambda \in F$.

Once we choose a basis (that is, coordinates) on V and W , we can identify $V \cong F^n$ and $W \cong F^m$, and then rewrite T as a map

$$T : F^n \longrightarrow F^m,$$

$$(x_1, \dots, x_n) \longmapsto (y_1, \dots, y_m),$$

where each output coordinate y_i is a linear combination of the input coordinates:

$$y_i = \sum_{j=1}^n a_{ij} x_j, \quad a_{ij} \in F.$$

The coefficients a_{ij} can be collected into a matrix, which encodes the linear transformation T .

Definition 2.1. Let $m, n \in \mathbb{N}$. The set of all $m \times n$ matrices with entries in a field F is

$$M_{m \times n}(F) := \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in F \right\}.$$

An element of $M_{m \times n}(F)$ is called an $m \times n$ matrix over F .

Given a matrix $A \in M_{m \times n}(F)$ and a column vector $x \in F^n$, we can define the product $Ax \in F^m$ by

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad Ax = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix},$$

where

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

This is exactly the situation above, with A representing the linear map T in the chosen bases.

Matrix multiplication

Definition 2.2. Let $A = (a_{ij}) \in M_{m \times n}(F)$ and $B = (b_{j\ell}) \in M_{n \times p}(F)$. The product AB is the $m \times p$ matrix

$$AB = C = (c_{i\ell}) \in M_{m \times p}(F),$$

whose entries are defined by

$$c_{i\ell} := \sum_{j=1}^n a_{ij} b_{j\ell}.$$

Observation 2.3. The matrix product AB is only defined when the number of columns of A equals the number of rows of B . In the notation above, A must be $m \times n$ and B must be $n \times p$. In that case, the product AB is an $m \times p$ matrix.

In terms of entries, the product looks like

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{np} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + \cdots + a_{1n}b_{n1} & \cdots & a_{11}b_{1p} + \cdots + a_{1n}b_{np} \\ \vdots & \ddots & \vdots \\ a_{m1}b_{11} + \cdots + a_{mn}b_{n1} & \cdots & a_{m1}b_{1p} + \cdots + a_{mn}b_{np} \end{pmatrix}.$$

Example 2.4. The zero matrix behaves as expected under multiplication. For instance,

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Example 2.5. Matrix multiplication is not commutative in general: it is usually not true that $AB = BA$.

Consider the two matrices

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

and $AB \neq BA$.

Matrices as a vector space

Proposition 2.6. For fixed m and n , the set $M_{m \times n}(F)$ of all $m \times n$ matrices with entries in F is a vector space over F under the following operations:

- *Addition:*

$$(a_{ij})_{m \times n} + (b_{ij})_{m \times n} := (a_{ij} + b_{ij})_{m \times n}.$$

- *Zero vector:* the zero matrix

$$0 := \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

satisfies $A + 0 = A = 0 + A$ for all $A \in M_{m \times n}(F)$.

- *Additive inverse:* for $A = (a_{ij})$, the matrix

$$-A := (-a_{ij})$$

satisfies $A + (-A) = 0 = (-A) + A$.

- *Scalar multiplication:* for $\lambda \in F$ and $A = (a_{ij})$,

$$\lambda A := (\lambda a_{ij}).$$

Proof. The verification is straightforward. For example, if $\lambda, \mu \in F$ and $A = (a_{ij})$, then

$$(\lambda\mu)A = ((\lambda\mu)a_{ij}) = (\lambda(\mu a_{ij})) = \lambda((\mu a_{ij})) = \lambda(\mu A),$$

so scalar multiplication is associative. Similarly,

$$\lambda(A + B) = \lambda(a_{ij} + b_{ij}) = (\lambda a_{ij} + \lambda b_{ij}) = \lambda A + \lambda B,$$

and

$$(\lambda + \mu)A = ((\lambda + \mu)a_{ij}) = (\lambda a_{ij} + \mu a_{ij}) = \lambda A + \mu A.$$

The remaining vector space axioms (such as commutativity and associativity of addition, existence of additive inverses, and the identity $1 \cdot A = A$) are checked entrywise and are left to the reader. $\square$

Remark 2.7. Unlike some other examples of vector spaces (such as the field F itself, or the polynomial ring $F[X]$), the space $M_{m \times n}(F)$ does not carry a natural matrix multiplication unless $m = n$. In general, the only multiplication available is the *composition-type* multiplication

$$M_{m \times n}(F) \times M_{n \times k}(F) \longrightarrow M_{m \times k}(F),$$

which corresponds to composing linear maps $F^k \rightarrow F^n \rightarrow F^m$. When $m = n = k$, this specialises to a multiplication

$$M_{n \times n}(F) \times M_{n \times n}(F) \longrightarrow M_{n \times n}(F),$$

making $M_{n \times n}(F)$ into a ring (and in fact an algebra over F , meaning that the multiplication is F -bilinear).

Systems of Linear Equations

Observation 2.8. Consider a matrix $A \in M_{m \times n}(F)$, a column vector of unknowns

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in F^n,$$

and a column vector

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in F^m.$$

The matrix equation

$$Ax = b$$

is equivalent to the system of m linear equations

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad \text{for all } i = 1, \dots, m,$$

where a_{ij} denotes the (i, j) -entry of A .

Notation 2.9. We often write the matrix equation in the compact form

$$Ax = b,$$

where

$$A \in M_{m \times n}(F), \quad x \in M_{n \times 1}(F), \quad b \in M_{m \times 1}(F).$$

The entries of x are the unknowns of the system, and the entries of b are the constants on the right-hand side.

Definition 2.10. Let $A \in M_{m \times n}(F)$ and $B \in M_{m \times p}(F)$. The *augmented matrix* associated to the pair (A, B) is the $m \times (n + p)$ matrix denoted by

$$(A \mid B),$$

obtained by placing the columns of B to the right of the columns of A .

Observation 2.11. In the context of a single system of linear equations $Ax = b$, it is common to take $B = b \in M_{m \times 1}(F)$ and to work with the augmented matrix $(A \mid b)$. This packages the coefficients of the unknowns and the constants on the right-hand side into a single matrix.

More formal properties of matrices

Bilinearity of the action of a matrix on a vector

Proposition 2.12. The map

$$M_{m \times n}(F) \times M_{n \times 1}(F) \longrightarrow M_{m \times 1}(F), \quad (A, v) \longmapsto Av,$$

is F -bilinear. That is, for all matrices $A, B \in M_{m \times n}(F)$, all column vectors $v, w \in M_{n \times 1}(F)$, and all scalars $\lambda, \mu \in F$, we have:

$$\begin{aligned} (\lambda A)v &= \lambda(Av), & A(\lambda v) &= \lambda(Av), \\ A(v + w) &= Av + Aw, & (A + B)v &= Av + Bv. \end{aligned}$$

Proof. Write $A = (a_{ij})$, $B = (b_{ij})$, $v = (v_j)$, and $w = (w_j)$. For the first identity,

$$((\lambda A)v)_i = \sum_j (\lambda a_{ij})v_j = \lambda \sum_j a_{ij}v_j = (\lambda(Av))_i,$$

so $(\lambda A)v = \lambda(Av)$. A similar calculation shows $A(\lambda v) = \lambda(Av)$:

$$(A(\lambda v))_i = \sum_j a_{ij}(\lambda v_j) = \lambda \sum_j a_{ij}v_j = (\lambda(Av))_i.$$

For additivity in the vector variable,

$$(A(v + w))_i = \sum_j a_{ij}(v_j + w_j) = \sum_j a_{ij}v_j + \sum_j a_{ij}w_j = (Av)_i + (Aw)_i,$$

so $A(v + w) = Av + Aw$.

For additivity in the matrix variable,

$$((A + B)v)_i = \sum_j (a_{ij} + b_{ij})v_j = \sum_j a_{ij}v_j + \sum_j b_{ij}v_j = (Av)_i + (Bv)_i,$$

so $(A + B)v = Av + Bv$. This proves bilinearity. $\square$

Corollary 2.13. For any fixed matrix $A \in M_{m \times n}(F)$, the map

$$T_A : F^n \longrightarrow F^m, \quad v \longmapsto Av,$$

is F -linear.

Proof. The bilinearity of $(A, v) \mapsto Av$ shows that for fixed A the map $v \mapsto Av$ is additive and homogeneous in v , which is precisely the definition of linearity. $\square$

Remark 2.14. Using the identification $F^n \cong M_{n \times 1}(F)$ and $F^m \cong M_{m \times 1}(F)$, we can also regard T_A as a linear map

$$M_{n \times 1}(F) \longrightarrow M_{m \times 1}(F),$$

still given by $v \mapsto Av$.

Matrix multiplication and columns

Proposition 2.15. Let $A \in M_{m \times n}(F)$ and let

$$B = (B_1 \ B_2 \ \cdots \ B_p) \in M_{n \times p}(F),$$

where B_j denotes the j -th column of B . Then

$$AB = (AB_1 \ AB_2 \ \cdots \ AB_p),$$

that is, the j -th column of AB is AB_j .

Proof. Let $C = AB$. The (i, j) -entry of C is

$$C_{ij} = (AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

On the other hand, the j -th column B_j is the vector whose k -th entry is B_{kj} , so the i -th entry of AB_j is

$$(AB_j)_i = \sum_k A_{ik} (B_j)_k = \sum_k A_{ik} B_{kj}.$$

Thus $(AB_j)_i = C_{ij}$ for all i , so the j -th column of C is AB_j , as claimed. $\square$

Remark 2.16. A completely analogous statement holds for rows. If

$$B = \begin{pmatrix} B_1 \\ \vdots \\ B_p \end{pmatrix}$$

is written in terms of its rows $B_1, \dots, B_p$, and A is a suitable matrix, then

$$BA = \begin{pmatrix} B_1 A \\ \vdots \\ B_p A \end{pmatrix},$$

so the i -th row of BA is $B_i A$.

Transpose of a matrix

Definition 2.17. Let $A \in M_{m \times n}(F)$. The *transpose* of A is the matrix $A^t \in M_{n \times m}(F)$ defined by

$$(A^t)_{ij} := A_{ji} \quad \text{for all } i, j.$$

Notation 2.18. If M is a matrix, we write M_{ij} for the (i, j) -entry of M .

Proposition 2.19. For any matrices A and B for which the product AB is defined, we have

$$(AB)^t = B^t A^t.$$

Proof. Let $C = AB$. Then

$$(AB)_{ij}^t = C_{ij}^t = C_{ji} = (AB)_{ji}.$$

By the definition of matrix multiplication,

$$(AB)_{ji} = \sum_k A_{jk} B_{ki}.$$

Now note that $B_{ki} = B_{ik}^t$ and $A_{jk} = A_{kj}^t$, so

$$\sum_k A_{jk} B_{ki} = \sum_k B_{ik}^t A_{kj}^t = (B^t A^t)_{ij}.$$

Thus $(AB)_{ij}^t = (B^t A^t)_{ij}$ for all i, j , so $(AB)^t = B^t A^t$. $\square$

Corollary 2.20. Matrix multiplication

$$M_{m \times n}(F) \times M_{n \times p}(F) \longrightarrow M_{m \times p}(F), \quad (A, B) \mapsto AB,$$

is F -bilinear.

Proof. Write $B = (B_1 \cdots B_p)$ in terms of its columns. By the proposition on columns, we have

$$(A, B) \mapsto AB = (AB_1 \cdots AB_p),$$

and each map $(A, B_j) \mapsto AB_j$ is bilinear by the earlier bilinearity result for $M_{m \times n}(F) \times M_{n \times 1}(F)$. Since matrix addition and scalar multiplication are done column by column, it follows that $(A, B) \mapsto AB$ is bilinear. $\square$

In particular, matrix multiplication is distributive over addition in each variable:

$$A(B + C) = AB + AC \quad \text{and} \quad (A + B)C = AC + BC$$

whenever the products are defined.

Identity matrices and associativity

Definition 2.21. The *identity matrix* of size n is the matrix

$$I_n \in M_{n \times n}(F), \quad I_n := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

A *scalar matrix* is a matrix of the form

$$\lambda I_n = \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda \end{pmatrix},$$

where $\lambda \in F$.

Observation 2.22. For any $A \in M_{m \times n}(F)$ we have

$$I_m A = A = A I_n.$$

More generally, for any scalar $\lambda \in F$,

$$(\lambda I_m) A = \lambda A = A (\lambda I_n),$$

whenever these products are defined.

Theorem 2.23. Matrix multiplication is associative. That is, whenever the products are defined,

$$(AB)C = A(BC).$$

Proof. Let $A \in M_{m \times n}(F)$, $B \in M_{n \times p}(F)$, and $C \in M_{p \times q}(F)$. Then

$$((AB)C)_{ij} = \sum_{\ell} (AB)_{i\ell} C_{\ell j} = \sum_{\ell} \left(\sum_k A_{ik} B_{k\ell} \right) C_{\ell j} = \sum_{k,\ell} A_{ik} B_{k\ell} C_{\ell j}.$$

On the other hand,

$$(A(BC))_{ij} = \sum_k A_{ik} (BC)_{kj} = \sum_k A_{ik} \left(\sum_{\ell} B_{k\ell} C_{\ell j} \right) = \sum_{k,\ell} A_{ik} B_{k\ell} C_{\ell j}.$$

Thus $((AB)C)_{ij} = (A(BC))_{ij}$ for all i, j , so $(AB)C = A(BC)$. $\square$

Corollary 2.24. For matrices A, B of compatible sizes and a column vector v , we have

$$A(Bv) = (AB)v.$$

In other words, matrix multiplication agrees with composition of the corresponding linear maps: if $T_B : v \mapsto Bv$ and $T_A : w \mapsto Aw$, then

$$T_A \circ T_B : v \mapsto A(Bv) = (AB)v.$$

Thus, at the level of matrices,

$$AB \quad \text{represents the composition} \quad T_A \circ T_B.$$

Transpose and some special types of matrices

Definition 2.25. Let $A \in M_{m \times n}(F)$. The *transpose* of A is the matrix

$$A^t \in M_{n \times m}(F)$$

defined by

$$(A^t)_{ij} := A_{ji} \quad \text{for all } i, j.$$

Example 2.26. The matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

satisfies $A^t = A$, so it is equal to its transpose.

Observation 2.27. For every matrix A , we have $(A^t)^t = A$. In particular, the transpose operation maps $M_{n \times n}(F)$ to itself: the transpose of a square matrix is again a square matrix of the same size.

Definition 2.28. Let $A \in M_{n \times n}(F)$ be a square matrix. We say that:

- A is *symmetric* if $A = A^t$.
- A is *antisymmetric* (or *skew-symmetric*) if $A = -A^t$.
- A is *orthogonal* (over $F = \mathbb{R}$; or more generally, *orthogonal/unitary* with respect to a suitable inner product) if

$$AA^t = I_n = A^t A.$$

Remark 2.29. Later we will prove that if $A, B \in M_{n \times n}(F)$ satisfy $AB = I_n$, then automatically $BA = I_n$ as well. In other words, for square matrices a right inverse is also a left inverse.

Definition 2.30. Let $A \in M_{n \times n}(F)$.¹

- We say that A is *upper triangular* if

$$A_{ij} = 0 \quad \text{whenever } i > j.$$

In other words, all entries strictly below the main diagonal are zero. A typical 2×2 example is

$$\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}.$$

- We say that A is *lower triangular* if

$$A_{ij} = 0 \quad \text{whenever } i < j.$$

That is, all entries strictly above the main diagonal are zero. A typical 2×2 example is

$$\begin{pmatrix} * & 0 \\ * & * \end{pmatrix}.$$

- We say that A is *diagonal* if

$$A_{ij} = 0 \quad \text{whenever } i \neq j.$$

Thus only the diagonal entries may be non-zero, for instance

$$\begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix}.$$

A quick application: Fibonacci numbers

Let $(F_n)_{n \geq 0}$ be the Fibonacci sequence, defined recursively by

$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2,$$

with initial values F_0 and F_1 .

We can package the recursion into a matrix equation:

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} F_{n-2} \\ F_{n-1} \end{pmatrix} = \begin{pmatrix} F_{n-1} \\ F_n \end{pmatrix}.$$

Iterating this matrix multiplication, we obtain

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^n \begin{pmatrix} F_0 \\ F_1 \end{pmatrix} = \begin{pmatrix} F_n \\ F_{n+1} \end{pmatrix}.$$

Computing the n -th power of a general 2×2 matrix directly is not very convenient. However, if we can write the matrix in diagonal form

$$D = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix},$$

¹In some places, one allows $A \in M_{m \times n}(F)$ to be nonsquare; the definitions here make sense in this generality. But we will not need to consider this in this course.

then

$$D^n = \begin{pmatrix} a^n & 0 \\ 0 & b^n \end{pmatrix},$$

which is easy to compute.

More generally, if a matrix A can be written as

$$A = P^{-1}DP$$

for some invertible matrix P and diagonal matrix D , then

$$A^n = (P^{-1}DP)^n = P^{-1}D^nP,$$

since matrix multiplication is associative and $P^{-1}P = I$.

In the case of the Fibonacci matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

one can show that

$$A^n = P^{-1} \begin{pmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{pmatrix} P$$

for a suitable invertible matrix P . Applying this to the initial vector $\begin{pmatrix} F_0 \\ F_1 \end{pmatrix}$, and choosing $F_0 = 0$, $F_1 = 1$, one eventually obtains the closed formula

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right],$$

known as Binet's formula. Note that the final formula for F_n can be written without any explicit reference to matrices, even though matrices provide a very efficient way to derive it.

Solving systems of equations

The 2×2 case

Consider the system

$$ax + by = e,$$

$$cx + dy = f,$$

which we can write in matrix form as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix}.$$

We can solve this system by eliminating one of the variables. Assume first that $c \neq 0$. Multiply the second equation by $-a/c$:

$$-\frac{a}{c}(cx + dy) = -\frac{a}{c}f.$$

Adding this to the first equation gives

$$ax + by - \frac{a}{c}(cx + dy) = e - \frac{a}{c}f,$$

so the x -terms cancel and we obtain

$$\left(b - \frac{ad}{c}\right)y = e - \frac{af}{c}.$$

If $ad - bc \neq 0$, then $b - \frac{ad}{c} = \frac{bc - ad}{c} \neq 0$, and hence

$$y = \frac{ec - af}{bc - ad}.$$

Substituting this expression for y back into one of the original equations gives

$$x = \frac{bf - de}{bc - ad}.$$

Thus

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} de - bf \\ af - ce \end{pmatrix}, \quad \text{provided } ad - bc \neq 0.$$

Remark 2.31. 1. Although we assumed $c \neq 0$ in the elimination step, the final formula still holds even if $c = 0$, provided $ad - bc \neq 0$.

2. If $ad - bc = 0$, then the left-hand sides $ax + by$ and $cx + dy$ are multiples of each other. In that situation, two things can happen:

- The system is *inconsistent*, for example

$$x + y = 1, \quad 2x + 2y = 1,$$

where the two equations contradict each other.

- The system is effectively equivalent to a single equation (or possibly has no equation at all, as in $0x + 0y = 0$). In this case there are infinitely many solutions if the field F is infinite.

Gaussian elimination

In the 2×2 example, we *eliminated* the variable x from one equation by adding a suitable multiple of the other equation, and then we performed back-substitution. This procedure generalises to larger systems and is called *Gaussian elimination*.

Consider, for instance,

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & -2 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}.$$

We can perform row operations to simplify the system. First use the first equation to eliminate x_1 from the second row:

$$R_2 \leftarrow R_2 - 2R_1,$$

giving

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}.$$

Next use the first equation to eliminate x_1 from the third row:

$$R_3 \leftarrow R_3 + R_1,$$

giving

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 8 \end{pmatrix}.$$

We then continue:

1. Use the first equation to eliminate x_1 from all other equations.
2. Use the second equation to eliminate x_2 from all later equations.
3. Continue in this way, then solve the resulting triangular system by back-substitution.

Continuing the example, subtract the second row from the third:

$$R_3 \leftarrow R_3 - R_2,$$

obtaining the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -4 & -2 \\ 0 & 0 & 5 & 10 \end{array} \right].$$

From the last row we get

$$5x_3 = 10 \quad \Rightarrow \quad x_3 = 2.$$

Then the second row gives

$$x_2 - 4x_3 = -2 \quad \Rightarrow \quad x_2 = -2 + 4 \cdot 2 = 6,$$

and the first row gives

$$x_1 + x_3 = 3 \quad \Rightarrow \quad x_1 = 3 - x_3 = 1.$$

This method works very well except in certain special situations.

Row operations and edge cases

Sometimes the top equation does not involve the first variable. For example,

$$\begin{pmatrix} 0 & 1 & * & | & * \\ 1 & * & * & | & * \end{pmatrix},$$

where $*$ denotes some (possibly non-zero) entry. In such a case we cannot use the first row to eliminate x_1 , because x_1 does not appear in that row.

Observation 2.32. In situations like this, we simply *permute* the rows so that an equation involving the next variable we want to eliminate appears first. Permuting equations does not change the solution set.

Observation 2.33. We are also allowed to multiply an equation by a non-zero scalar. For instance,

$$3x_3 + \cdots = 6 \quad \Longleftrightarrow \quad x_3 + \cdots = 2.$$

In matrix form, this corresponds to multiplying a row by a non-zero scalar to make its leading non-zero coefficient equal to 1.

After such operations, a typical matrix obtained by Gaussian elimination has a staircase, or “echelon” form, with 1’s in the corner, looking like:

$$\begin{pmatrix} 1 & \cdots & * & * & \cdots & * & * & * & \cdots & * \\ 0 & \cdots & 0 & 1 & \cdots & * & * & * & \cdots & * \\ 0 & \cdots & 0 & 0 & \cdots & 0 & 1 & * & \cdots & * \\ \vdots & & & & & & & & & \end{pmatrix},$$

where one can have some zero rows at the bottom, and any number of nonzero rows; in the example we have at least three, but there can be fewer or more.

Row echelon form and pivots

Definition 2.34. A matrix is said to be in *row echelon form* (often abbreviated *REF*) if:

- Every non-zero row has a first non-zero entry equal to 1 (this is called the *leading* or *pivot* entry).
- As we move down the rows, the position of the first non-zero entry moves strictly to the right. That is, the pivot in each subsequent non-zero row appears in a later column than in the row above.
- Any rows consisting entirely of zeros appear (if at all) at the bottom of the matrix.

Definition 2.35. In a matrix in row echelon form, a column that contains a pivot (that is, a leading 1 of some row) is called a *pivot column*. The variables corresponding to pivot columns are called *pivot variables*.

Solving a system in row echelon form

Once a system has been reduced to row echelon form, we can solve it starting from the last non-zero row and working upwards, expressing pivot variables in terms of the remaining variables.

Example 2.36. Consider

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 8 \end{bmatrix}.$$

From the last row we obtain

$$x_5 = 8.$$

From the second row,

$$x_4 + 6x_5 = 7 \quad \Rightarrow \quad x_4 = 7 - 6 \cdot 8 = -41.$$

From the first row,

$$x_2 + 2x_3 + 3x_4 + 4x_5 = 5.$$

Substituting x_4 and x_5 we find

$$x_2 = 5 - 2x_3 - 3x_4 - 4x_5 = 5 - 2x_3 - 3(-41) - 4 \cdot 8 = -2x_3 + 96.$$

Here x_1 and x_3 do not appear as pivot variables; they are *free variables*. The pivot variables are x_2 , x_4 , and x_5 , which we have expressed in terms of the free variables.

Notation 2.37. The number of pivot variables (equivalently, the number of pivot columns) is called the *rank* of the augmented matrix $(A \mid b)$.

The Gaussian elimination algorithm

Formally, Gaussian elimination proceeds as follows:

1. If necessary, permute the rows so that in the first non-zero column, the top row has a non-zero entry:

$$\begin{bmatrix} 0 & \boxed{\alpha} & * \\ 0 & * & * \end{bmatrix}, \quad \alpha \neq 0.$$

2. Rescale the first row so that its leading entry (here α) becomes 1.
3. Add suitable multiples of the first row to all other rows to eliminate that leading variable from the other equations, producing zeros below the pivot:

$$\sim \begin{bmatrix} 0 & 1 & * \\ 0 & 0 & * \end{bmatrix}.$$

4. Ignore (or delete) the first row and the first pivot column, and repeat the procedure on the remaining rows and columns.

This algorithm produces a matrix in row echelon form.

Solving from a row echelon form

Once we have a row echelon form, we proceed as follows:

- Identify the *free* variables: these correspond to columns which are not pivot columns.
- Starting from the last pivot row and moving upwards, solve each pivot variable in terms of the free variables (this is back-substitution).

Gauss–Jordan elimination and reduced row echelon form

Gauss–Jordan elimination is a refinement of Gaussian elimination that removes the need for back-substitution. After obtaining a row echelon form, we continue with additional row operations to simplify the matrix further: we use the pivot rows to eliminate the entries above each pivot as well.

Definition 2.38. A matrix is in *reduced row echelon form* (often abbreviated *RREF*) if:

- It is in row echelon form.

- Every pivot entry is equal to 1.
- Each pivot is the only non-zero entry in its column.

Equivalently, the pivot columns are standard basis vectors

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix},$$

with a single 1 and all other entries zero. In this form, solving the system requires no back-substitution: each pivot equation directly expresses a pivot variable in terms of the free variables.

Below we will prove that the reduced row echelon form of a matrix is *unique*. As a consequence, the number of pivot columns (the rank) is independent of the sequence of row operations used.

Remark 2.39. In practice, during Gauss–Jordan elimination, the only real choices we make are which row permutations to apply and in what order to perform elimination. The uniqueness theorem says that, regardless of the choices, the final reduced row echelon form is always the same.

Remark 2.40. Alternatively, we can deduce that the rank is well-defined by looking at the number of free variables in the solution set. The general solution of a consistent system depends on one parameter for each free variable, so the number of free variables is determined by the solution set and does not depend on the row operations used. Since

$$\#\text{free variables} = n - \#\text{pivot variables},$$

where n is the total number of variables, the number of pivot variables (and hence the rank) is also uniquely determined.

To solve a system in reduced row echelon form, we proceed exactly as in the row echelon case, but we do not need back-substitution, and we may solve the pivot equations in any convenient order.

Example 2.41. Consider the reduced row echelon matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 7 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$

as the coefficient matrix (including the right-hand side as the last column) for a system in the variables $x_1, \dots, x_5$. The equations are

$$x_2 + 7x_5 = 2, \quad x_3 - x_5 = -1, \quad x_4 + 2x_5 = -4.$$

Solving for the pivot variables we obtain

$$x_2 = 2 - 7x_5, \quad x_3 = -1 + x_5, \quad x_4 = -4 - 2x_5.$$

The variables x_1 and x_5 are free.

Row equivalence and uniqueness of reduced row echelon form

Definition 2.42. Let $A, B \in M_{m \times n}(F)$. We say that A and B are *row-equivalent* if B can be obtained from A by a finite sequence of the following *elementary row operations*:

- permuting two rows,
- multiplying a row by a non-zero scalar,
- adding a scalar multiple of one row to another row.

Theorem 2.43. Every matrix is row-equivalent to a matrix in reduced row echelon form.

Proof. This is precisely what the Gauss–Jordan elimination algorithm achieves. $\square$

Theorem 2.44. A reduced row echelon form augmented matrix $(A \mid b)$ is inconsistent if and only if the last column is a pivot column.

Proof. If the last column is a pivot column, then there is a row of the form

$$(0 \ 0 \ \cdots \ 0 \mid 1),$$

which corresponds to the equation $0 = 1$, so the system is inconsistent.

Conversely, suppose the system is inconsistent. In reduced row echelon form, an inconsistency must appear as a row $0 \cdots 0 \mid c$ with $c \neq 0$, and the last column must then contain a pivot in that row. Hence the last column is a pivot column. $\square$

In the consistent case, once we know which columns are pivot and which are free, the general solution is uniquely determined. Two different reduced row echelon form matrices with the same pattern of pivot and free columns would produce different solution sets, so this pattern is intrinsic to the matrix.

Pivot columns and solution sets

Suppose the general solution of a consistent system can be written as

$$x_i = c_i + a_{i1}v_{j_1} + \cdots + a_{ip}v_{j_p},$$

where $v_{j_1}, \dots, v_{j_p}$ are the free variables and c_i is a constant (possibly zero). Setting all the free variables to zero recovers $x_i = c_i$. Varying one free variable at a time (for instance setting $(v_{j_1}, \dots, v_{j_p}) = e_r$) allows us to recover the coefficients a_{ir} .

Observation 2.45. For a consistent system $Ax = b$ with $A \in M_{m \times n}(F)$, the i -th column is a pivot column if and only if the variable x_i is uniquely determined by the variables $x_{i+1}, \dots, x_n$ (equivalently, by the free variables). Thus the set of pivot columns is uniquely determined by the solution set.

Theorem 2.46. (i) If $(A \mid b)$ is row-equivalent to $(A' \mid b')$, then the two systems $Ax = b$ and $A'x = b'$ have the same set of solutions.

(ii) Conversely, if $(A \mid b)$ and $(A' \mid b')$ are consistent augmented matrices in $M_{m \times (n+1)}(F)$ and they have the same set of solutions, then they are row-equivalent.

- (iii) For any matrix A , there is a *unique* reduced row echelon form matrix B that is row-equivalent to A .

Proof. (i) We check that each elementary row operation preserves the solution set.

- Permuting rows does not change the set of equations, only their order.
 - Multiplying a row by a non-zero scalar λ replaces an equation of the form $\lambda f(x) = 0$ by $f(x) = 0$, which has the same solutions since $\lambda \neq 0$.
 - Adding a multiple of one row to another replaces an equation $f_j(x) = 0$ by $f_j(x) + \lambda f_i(x) = 0$, where $f_i(x) = 0$ is another equation in the system. If $f_i(x) = 0$ and $f_j(x) = 0$ both hold, then so does $f_j(x) + \lambda f_i(x) = 0$, and conversely, from $f_j(x) + \lambda f_i(x) = 0$ and $f_i(x) = 0$ we recover $f_j(x) = 0$. Thus the set of common solutions is unchanged.
- (ii) If $(A \mid b)$ and $(A' \mid b')$ are consistent and have the same solutions, then their reduced row echelon forms must also have the same solutions. But a reduced row echelon form is determined by its solution set, so the reduced forms must be equal. Therefore $(A \mid b)$ and $(A' \mid b')$ are row-equivalent.
- (iii) Let B and B' be two reduced row echelon form matrices that are row-equivalent to A . Then B and B' are row-equivalent to each other, hence, by (i), define systems with the same solution set. By (ii), B and B' must be row-equivalent and in reduced row echelon form; but a reduced row echelon form is determined by its solution set, so $B = B'$. Hence there is a unique reduced row echelon form matrix row-equivalent to A .

□

Inverting matrices

Right inverses and solving multiple systems

Definition 2.47. Let $A \in M_{m \times n}(F)$. A matrix $C \in M_{n \times m}(F)$ is called a *right inverse* of A if

$$AC = I_m.$$

Similarly, a matrix $B \in M_{n \times m}(F)$ is called a *left inverse* if $BA = I_n$.

Observation 2.48. The equation

$$AC = I_m$$

holds if and only if the columns $C_1, \dots, C_m$ of C satisfy

$$AC_i = e_i \quad \text{for all } i = 1, \dots, m,$$

where e_i is the i -th standard basis vector of F^m . Thus finding a right inverse amounts to solving the systems $Ax = e_i$ for each i .

Observation 2.49. To solve several systems

$$Ax = B_1, \quad Ax = B_2, \quad \dots, \quad Ax = B_k,$$

we do *not* need to run Gaussian elimination k times. Instead, we form the augmented matrix $(A \mid B_1 \cdots B_k)$ and perform Gauss–Jordan elimination once:

$$(A \mid B_1 \cdots B_k) \xrightarrow{\text{GJ}} (R \mid C_1 \cdots C_k),$$

where R is in reduced row echelon form and each C_i is the transformed right-hand side. Then solving $Ax = B_i$ is equivalent to solving $Rx = C_i$ for each i .

Example 2.50. Consider

$$\left(\begin{array}{c|cc} 2 & 1 & 0 \\ 1 & 0 & 1 \end{array} \right).$$

Performing row operations, we obtain

$$\left(\begin{array}{c|cc} 2 & 1 & 0 \\ 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{c|cc} 1 & \frac{1}{2} & 0 \\ 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{c|cc} 1 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{array} \right).$$

Interpreting this, we see that the 2×1 matrix

$$A = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

does *not* have a right inverse: there is no 1×2 matrix C such that $AC = I_2$, because A represents a linear map from F^2 to F and cannot be bijective. On the other hand, the row matrix

$$(2 \quad 1)$$

does have (many) right inverses: solving

$$(2 \quad 1) \begin{pmatrix} a \\ b \end{pmatrix} = 1$$

gives $2a + b = 1$, which has infinitely many solutions if $2 \neq 0$ in F , for example $(a, b) = (0, 1)$ or $(a, b) = (1/2, 0)$.

Constraints on left and right inverses

Claim 2.51. If $AC = I_m$ has a solution (that is, if A has a right inverse), then $m \leq n$, and the rref form matrix of A has no zero rows.

Proof. For the first statement, suppose by contradiction that $m > n$. Consider the augmented matrix $(A \mid I_m)$ of size $m \times (n + m)$ and apply Gauss–Jordan elimination:

$$(A \mid I_m) \sim (R \mid C),$$

with R in reduced row echelon form. Since $m > n$, the matrix R must have at least one zero row.

If $AC = I_m$ for some C , then the system $AX = I_m$ is consistent, so the corresponding system $RX = C$ is also consistent. Therefore C must have a zero row in the same position. But applying Gauss–Jordan elimination to (I_m) alone would produce I_m , which has no zero rows. This contradicts the uniqueness of the reduced row echelon form. Hence $m \leq n$.

The same proof in the preceding paragraph shows that, if $\text{rref}(A)$ has a zero row, then A has no right inverse. $\square$

Corollary 2.52. If $A \in M_{n,n}$ is a square matrix and A has a right inverse, then $\text{rref}(A) = I$.

Proof. We saw that the rref form of A has no zero rows. Since it is square, it must be the identity. $\square$

Transposing, we obtain:

Corollary 2.53. If $BA = I_n$ has a solution (that is, if A has a left inverse), then $m \geq n$.

Corollary 2.54. If $A \in M_{n,n}$ is a square matrix and A has a left inverse, then $\text{rref}(A) = I$.

Corollary 2.55. If A has both a left inverse and a right inverse, then A must be square (that is, $m = n$). Moreover, if A is square and $AB = I$ and $CA = I$, then $B = C$; this common matrix is the *inverse* A^{-1} , and it is unique.

Proof. This follows immediately from the preceding and the following problem sheet exercise: if $AB = I$ and $CA = I$, then $B = IB = (CA)B = C(AB) = CI = C$. $\square$

Corollary 2.56. If A is invertible, then

$$(A \mid I) \xrightarrow{\text{Gauss-Jordan}} (I \mid A^{-1})$$

(see Theorem 2.6.2).

Proof. As we saw, A has to be square and $\text{rref}(A) = I$. Let

$$(A \mid I) \sim (I \mid C)$$

be the result of Gauss-Jordan elimination. So $AB = I$ is equivalent to $IB = C$, thus $A^{-1} = C$. $\square$

Remark 2.57. On Problem Sheet 2 you will see that each elementary row operation can be realised as left multiplication by an *elementary matrix* E , that is,

$$A \mapsto EA.$$

This gives another proof that

$$(A \mid I) \xrightarrow{\text{row operations}} (I \mid A^{-1}) :$$

if $E_k \cdots E_2 E_1 A = I$, then

$$E_k \cdots E_2 E_1 = A^{-1}.$$

It also proves the converse: if $(A \mid I) \mapsto (I \mid C)$ by row operations, then $E_k \cdots E_1 A = I$, and since E_i are invertible, we get that $A = E_1^{-1} \cdots E_k^{-1}$ is invertible.

Examples of invertibility

Example 2.58. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

We claim that A is not invertible. Performing Gaussian elimination, we find

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 6 & 6 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix},$$

so the reduced row echelon form is not the identity. A more systematic reduction gives

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -6 & -12 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix},$$

again showing that the reduced row echelon form has a zero row and is not the identity. Therefore A is not invertible.

Example 2.59. Consider the matrix

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 3 & 6 & 8 & 0 & 1 & 0 \\ 2 & 6 & 6 & 0 & 0 & 1 \end{array} \right).$$

Applying Gauss–Jordan elimination, we obtain

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 0 & 2 & -3 & 1 & 0 \\ 0 & 2 & 0 & -2 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{3}{2} & \frac{1}{2} & 0 \end{array} \right)$$

and finally

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{15}{2} & -\frac{3}{2} & -1 \\ 0 & 1 & 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{3}{2} & \frac{1}{2} & 0 \end{array} \right).$$

Thus the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 6 & 8 \\ 2 & 6 & 6 \end{pmatrix}$$

is invertible, and its inverse is the 3×3 matrix on the right-hand side.

Example 2.60 (General 2×2 inverse). Consider the augmented matrix

$$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right).$$

Assume first that $a \neq 0$. Then

$$\sim \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & d - \frac{cb}{a} & -\frac{c}{a} & 1 \end{array} \right) = \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array} \right).$$

If $ad - bc \neq 0$, we can rescale the second row to obtain

$$\sim \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right).$$

Thus

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Even if $a = 0$, as long as $ad - bc \neq 0$ we can permute the rows to ensure that the first pivot is non-zero and obtain the same formula. Indeed, $ad - bc \neq 0$ implies that at least one of a or c is non-zero.

Corollary 2.61. A 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is invertible if and only if $ad - bc \neq 0$. In that case the inverse is given by

$$A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Final comments: determinants

There exists a determinant function

$$\det : M_{n \times n}(F) \longrightarrow F$$

which we will study in detail next term. For 2×2 matrices it is given by

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

A fundamental fact is that

$$\det A \neq 0 \iff A \text{ is invertible.}$$

- One can express the inverse of an invertible matrix A in terms of determinants:

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A),$$

where the *adjugate* $\operatorname{adj}(A)$ is the matrix whose (i, j) -entry is

$$\operatorname{adj}(A)_{ij} = (-1)^{i+j} \det(\text{the matrix obtained from } A \text{ by deleting row } j \text{ and column } i).$$

For example, in the 2×2 case this gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

in agreement with our earlier calculation.

- In practice, this determinant-based formula is *slower* than using Gauss–Jordan elimination, because it requires computing many determinants.
- One can solve $Ax = b$ using $x = A^{-1}b$ when A is invertible, for example

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix}, \quad ad - bc \neq 0,$$

gives

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} e \\ f \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} de - bf \\ -ce + af \end{pmatrix}.$$

- However, in general it is faster to solve $Ax = b$ directly by applying Gaussian (or Gauss–Jordan) elimination to the augmented matrix $(A \mid b)$, rather than first computing A^{-1} and then multiplying by b . Moreover, the augmented-matrix method works even when A is not invertible (that is, when $\det A = 0$).

We will not develop the theory of determinants in detail this term; that will be a major topic next term. You are welcome to use determinants if you are familiar with them, but they are not required for this term and will not be necessary for the January test.

3 Abstract vector spaces: linear independence, spanning, bases, and dimension; and subspaces associated to matrices

Let V be a vector space over a field F .

Linear combinations, independence, span, basis

Definition 3.1. For $n \geq 1$, given vectors $v_1, \dots, v_n \in V$ and scalars $\lambda_1, \dots, \lambda_n \in F$, an expression of the form

$$\lambda_1 v_1 + \dots + \lambda_n v_n$$

is called a *linear combination* of $v_1, \dots, v_n$.

Remark 3.2. Caution: Note that n is a natural number and in particular *all linear combinations are finite*. In analysis you can make sense of infinite sums, but these are not called linear combinations, rather limits of linear combinations.

Definition 3.3. An n -tuple $(v_1, \dots, v_n) \in V^n$, for $n \in \mathbb{N}$, is said to be *linearly dependent* if there exist scalars $\lambda_1, \dots, \lambda_n \in F$, not all zero, such that

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0.$$

If no such non-trivial relation exists, we say that $(v_1, \dots, v_n)$ (or the set $\{v_1, \dots, v_n\}$) is *linearly independent*. In particular, the empty tuple $()$ is also linearly independent.

Definition 3.4. Given vectors $v_1, \dots, v_n \in V$, for $n \geq 1$, the *span* of these vectors is the subset

$$\text{Span}(v_1, \dots, v_n) = \{\lambda_1 v_1 + \dots + \lambda_n v_n : \lambda_1, \dots, \lambda_n \in F\} \subseteq V.$$

By convention, the span of the empty tuple is the set containing zero: $\text{Span}(()) := \{0\}$.

Definition 3.5. Let $v_1, \dots, v_n \in V$.

- The n -tuple $(v_1, \dots, v_n)$ is said to be *spanning* if

$$\text{Span}(v_1, \dots, v_n) = V.$$

- The n -tuple $(v_1, \dots, v_n)$, for $n \in \mathbb{N}$, is called a *basis* of V if it is both linearly independent and spanning.

Main goal: Prove that any two bases of V have the same size. We will call this common size the *dimension* of V , denoted $\dim V$.

Remark 3.6. In this course we will only deal with finite tuples. On the other hand, when dealing with sets instead of tuples, as we will do shortly, we will occasionally allow ourselves to work with linear independence or spanning properties of infinite sets. This is mostly for convenience, so as to not have to develop notation for infinite tuples.

Basic examples

- The empty tuple $()$ is linearly independent and spans the zero vector space $\{0\}$ (by definition).
- The single-element tuple (0) is linearly dependent and spans $\{0\}$.
- The tuple (v, v) is linearly dependent for any $v \in V$. More generally, if $(v_1, \dots, v_n)$ contains a repeated vector $v_i = v_j$ for some $i \neq j$, then it is linearly dependent.
- In F^n , the standard coordinate vectors $e_1, \dots, e_n$ form a basis. Indeed,

$$\lambda_1 e_1 + \dots + \lambda_n e_n = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix},$$

so every vector has a unique expansion in this basis.

We could prove the main goal (that all bases have the same size) rather quickly, but instead we first look at several examples and basic results to become comfortable with these notions.

Exercises on independence, spanning, and subspaces

In the following examples, all vector spaces are over the indicated field.

- Decide which pairs or triples are *linearly independent*:
 - $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in F^2 .
 - $\begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ in F^2 .
 - $x^2 + x + 1, 3x + 4, 5$ in $\mathbb{R}[x]$ over $\mathbb{R}$.
 - $1, i$ in $\mathbb{C}$ regarded as a vector space over $\mathbb{R}$.
 - $\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ in F^2 .
 - $\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 700 \\ 10^{120} \\ 2024 \end{pmatrix}$ in F^3 .
 - $1, i$ in $\mathbb{C}$ regarded as a vector space over $\mathbb{C}$.
- Decide which of these subsets are *spanning* for the relevant vector spaces.
- Decide which of the following subsets of F^2 are *subspaces*:
 - $\left\{ \begin{pmatrix} a \\ 0 \end{pmatrix} : a \in F \right\}$.
 - $\left\{ \begin{pmatrix} a \\ a \end{pmatrix} : a \in F \right\}$.
 - $F \setminus \{13\} \subseteq F$.
 - $\left\{ \begin{pmatrix} a \\ a \end{pmatrix} : a \in F \right\} \cup \left\{ \begin{pmatrix} a \\ -a \end{pmatrix} : a \in F \right\}$.

Intersections and sums of subspaces

Proposition 3.7 (cf. Thm 3.2.6). Let $U, W \subseteq V$ be subspaces. Then:

- $U \cap W$ is a subspace of V .
- $U + W := \{u + w : u \in U, w \in W\}$ is a subspace of V .
- In general, $U \cup W$ is *not* a subspace.

Proof. For $U \cap W$:

- If $v_1, v_2 \in U \cap W$, then $v_1 + v_2 \in U$ and $v_1 + v_2 \in W$, hence $v_1 + v_2 \in U \cap W$.
- If $v \in U \cap W$ and $\lambda \in F$, then $\lambda v \in U$ and $\lambda v \in W$, so $\lambda v \in U \cap W$.
- Since $0 \in U$ and $0 \in W$, we have $0 \in U \cap W$.

For $U + W$:

- If $u_1, u_2 \in U$ and $w_1, w_2 \in W$, then

$$(u_1 + w_1) + (u_2 + w_2) = (u_1 + u_2) + (w_1 + w_2) \in U + W.$$

- For $\lambda \in F$ and $u \in U, w \in W$, we have

$$\lambda(u + w) = \lambda u + \lambda w \in U + W.$$

Thus both $U \cap W$ and $U + W$ are subspaces of V . □

Generalisation. Let $(U_i)_{i \in I}$ be a family of subspaces of V .

- The intersection

$$\bigcap_{i \in I} U_i$$

is a subspace of V .

- The sum

$$\sum_{i \in I} U_i := \{u_{i_1} + \cdots + u_{i_n} : n \geq 1, i_1, \dots, i_n \in I, u_{i_j} \in U_{i_j}\}$$

is also a subspace of V .

If some indices appear more than once, we may combine equal indices using closure of each U_i under addition, so it suffices to consider sums with distinct indices.

Proof. The proofs are the same as in the case of two subspaces: intersections are closed under sums and scalar multiples coordinatewise, and sums stay closed by grouping all U_i -parts together. □

Span of sets

Given any subset $X \subseteq V$, we define its span by

$$\text{Span}(X) := \{\lambda_1 x_1 + \cdots + \lambda_n x_n : n \geq 1, \lambda_i \in F, x_i \in X\} \cup \{0\}.$$

Remark 3.8. We may always assume the vectors $x_1, \dots, x_n$ are distinct: if some x_i is repeated, we combine the corresponding coefficients.

Observation 3.9. For finitely many vectors $u_1, \dots, u_n$ we can write

$$\text{Span}(u_1, \dots, u_n) = \text{Span}(\{u_1, \dots, u_n\}).$$

Proposition 3.10. For any subset $X \subseteq V$, the set $\text{Span}(X)$ is a subspace of V .

Proof. Let $v = \lambda_1 x_1 + \cdots + \lambda_m x_m$ and $w = \mu_1 y_1 + \cdots + \mu_n y_n$ be elements of $\text{Span}(X)$, with $x_i, y_j \in X$ and $\lambda_i, \mu_j \in F$. Then

$$v + w = \lambda_1 x_1 + \cdots + \lambda_m x_m + \mu_1 y_1 + \cdots + \mu_n y_n \in \text{Span}(X),$$

and for $\alpha \in F$,

$$\alpha v = \alpha \lambda_1 x_1 + \cdots + \alpha \lambda_m x_m \in \text{Span}(X).$$

Thus $\text{Span}(X)$ is closed under addition and scalar multiplication, and contains 0, so it is a subspace. $\square$

Proposition 3.11. If $W \subseteq V$ is a subspace and $X \subseteq V$ is a subset, then $X \subseteq W$ implies $\text{Span}(X) \subseteq W$.

Proof. If $x_1, \dots, x_n \in X$, then $x_1, \dots, x_n \in W$, and $\lambda_i x_i \in W$ for all i as W is a subspace. Then by applying closure under addition and induction on n , we have

$$\lambda_1 x_1 + \cdots + \lambda_n x_n \in W. \quad \square$$

The two previous propositions have the following useful consequence:

Corollary 3.12. If $X \subseteq \text{Span}(Y)$, then $\text{Span}(X) \subseteq \text{Span}(Y)$.

Proof. This is Proposition 3.11 using that $W := \text{Span}(Y)$ is a subspace, by Proposition 3.10. $\square$

Remark 3.13. The two propositions imply the following characterisation of span: $\text{Span}(X) \subseteq V$ is the intersection of all subspaces of V containing the subset $X \subseteq V$.

Next we consider unions of sets and relate them to sums of their spans:

Observation 3.14. Even if U_1 and U_2 are subspaces of V , their union $U_1 \cup U_2$ need not be a subspace.

Exercise 3.15. Show that $U_1 \cup U_2$ is a subspace of V only in the trivial cases $U_1 \subseteq U_2$ or $U_2 \subseteq U_1$.

For subspaces U_1, U_2 we have

$$\text{Span}(U_1 \cup U_2) = \{u_1 + u_2 : u_1 \in U_1, u_2 \in U_2\} = U_1 + U_2.$$

More generally,

$$\text{Span}\left(\bigcup_{i \in I} U_i\right) = \sum_{i \in I} U_i.$$

Row space and column space

Let $A \in M_{m \times n}(F)$ be a matrix,

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

- The *row space* of A is

$$\text{rowspace}(A) := \text{Span}(\text{rows of } A) \subseteq M_{1 \times n}(F) \cong F^n.$$

- The *column space* of A is

$$\text{colspace}(A) := \text{Span}(\text{columns of } A) \subseteq M_{m \times 1}(F) \cong F^m.$$

Example 3.16. • If $A = 0$, then

$$\text{rowspace}(A) = \{0\}, \quad \text{colspace}(A) = \{0\}.$$

- If $A = I_n$, then

$$\text{rowspace}(A) \cong F^n, \quad \text{colspace}(A) \cong F^n.$$

- If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix},$$

then one can check (for example by row and column operations) that

$$\text{rowspace}(A) = \{(a + b, 2a + b, 3a + b) : a, b \in F\},$$

$$\text{colspace}(A) = \left\{ \begin{pmatrix} a + b \\ 3a + b \\ 6a + b \end{pmatrix} : a, b \in F \right\}.$$

Exercise 3.17. Verify the descriptions of the row space and column space in the previous example.

Extending and shrinking linearly independent and spanning sets

Definition 3.18. A subset $X \subseteq V$ is said to be *linearly dependent* if there exist distinct elements $x_1, \dots, x_n \in X$ and scalars $\lambda_1, \dots, \lambda_n \in F$, not all zero, such that

$$\lambda_1 x_1 + \cdots + \lambda_n x_n = 0.$$

Otherwise X is *linearly independent*.

Remark 3.19. It is equivalent to require in the definition that $n \geq 1$ and all $\lambda_i \neq 0$, since we may simply omit any x_i whose coefficient is 0.

Proposition 3.20. An n -tuple $(v_1, \dots, v_n)$ is linearly independent if and only if the vectors $v_1, \dots, v_n$ are distinct and the set $\{v_1, \dots, v_n\}$ is linearly independent.

Proof. If $v_i = v_j$ for some $i \neq j$, then

$$0v_1 + \cdots + 0v_i + v_i + 0v_{i+1} + \cdots + 0v_{j-1} - v_j + v_{j+1} + \cdots + v_n = 0,$$

with a non-trivial choice of coefficients, so the tuple is linearly dependent.

Conversely, if $v_1, \dots, v_n$ are distinct and the tuple $(v_1, \dots, v_n)$ is linearly dependent, then there exist scalars $\lambda_1, \dots, \lambda_n$, not all zero, such that

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = 0.$$

This shows that the set $\{v_1, \dots, v_n\}$ is linearly dependent. Conversely, if

$$\lambda_1 v_{i_1} + \cdots + \lambda_m v_{i_m} = 0$$

is a non-trivial dependence among distinct elements of the set, we can extend this by inserting zeros in the missing positions to obtain a linear dependence of the entire tuple. Thus the two notions coincide. $\square$

Observations. If $X \subseteq Y \subseteq V$ and X is linearly dependent, then Y is also linearly dependent.

Equivalently, if Y is linearly independent, then every subset $X \subseteq Y$ is linearly independent.

Similarly, if $X \subseteq Y$ and X is spanning, then Y is also spanning.

• As a consequence:

- Every subset of a basis is linearly independent (but need not span).
- Every superset of a basis is spanning (but need not be linearly independent).

Our main goal generalises to the following statement:

If $X \subseteq V$ is linearly independent and $Y \subseteq V$ is spanning, then

$$|X| \leq |Y|.$$

In words: linearly independent sets cannot be “too large” and spanning sets cannot be “too small”.

First lemmas: extending and shrinking

Lemma 3.21. (i) Let $X \subseteq V$ be linearly independent and let $v \in V \setminus \text{Span}(X)$. Then $X \cup \{v\}$ is also linearly independent.

(ii) Let $X \subseteq V$ be spanning and let $x \in X$ be such that

$$x \in \text{Span}(X \setminus \{x\}).$$

Then $X \setminus \{x\}$ is still spanning.

Proof. (i) Suppose

$$\lambda_1 x_1 + \cdots + \lambda_n x_n + \lambda v = 0,$$

with $x_i \in X$. If $\lambda = 0$, then linear independence of X implies $\lambda_1 = \cdots = \lambda_n = 0$. If $\lambda \neq 0$, then

$$v = -\lambda^{-1}(\lambda_1 x_1 + \cdots + \lambda_n x_n) \in \text{Span}(X),$$

which contradicts the assumption $v \notin \text{Span}(X)$. Hence all coefficients are zero, and $X \cup \{v\}$ is linearly independent.

(ii) Suppose X is spanning and $x \in X$ satisfies

$$x = \lambda_1 x_1 + \cdots + \lambda_n x_n$$

for some $x_1, \dots, x_n \in X \setminus \{x\}$. Any linear combination

$$\mu_1 y_1 + \cdots + \mu_m y_m + \mu x, \quad y_i \in X \setminus \{x\},$$

can be rewritten using the above relation as a linear combination of elements of $X \setminus \{x\}$ only. Hence $\text{Span}(X \setminus \{x\}) = \text{Span}(X) = V$, so $X \setminus \{x\}$ is still spanning. $\square$

Remark 3.22. The first part can also be interpreted as *a linearly independent set which is not spanning is properly contained in a larger linearly independent set*. Contrapositively, the *maximal* linearly independent sets (i.e., ones not contained in larger linearly independent sets) are bases. We give the same proof of this in Proposition 3.32.

Conversely, if $x \in \text{Span}(X)$, and $x \notin X$, then $X \cup \{x\}$ is not linearly independent, writing the linear relation $x = 0 + \sum_{i=1}^n \lambda_i x_i$ for some $n \geq 0$ and some distinct $x_i \in X$. As a result, *maximal linearly independent subsets are the same as bases*.

We can easily modify the proof of (ii) to show that any spanning set which is linearly dependent properly contains another spanning set. The contrapositive says that every *minimal* spanning set (one that does not properly contain another spanning set) is a basis. We will give a slightly different proof of this in Proposition 3.29.

Conversely, if a spanning set X properly contains another spanning set X' , then writing an element of $X \setminus X'$ as a linear combination of elements of X' yields a linear dependence. So *minimal spanning sets are the same as bases*.

This is valid even for infinite-dimensional vector spaces V . For finite-dimensional vector spaces we will prove later that there is some $\dim V \in \mathbb{N}$ such that linearly independent sets have size at most $\dim V$, with equality if and only if it is a basis, and similarly spanning sets have size at least $\dim V$, with equality if and only if it is a basis. This gives another proof of the preceding paragraph, which proves more (but also requires more, and has restricted generality).

Steinitz Exchange Lemma

Lemma 3.23 (Steinitz Exchange Lemma, Thm 3.6.1). Let $X \subseteq V$ and let $v \in \text{Span}(X)$ with $v \neq 0$. Suppose

$$v = \lambda_1 x_1 + \cdots + \lambda_n x_n,$$

for distinct $x_1, \dots, x_n \in X$ and all $\lambda_i \neq 0$. Then for any index i ,

$$\text{Span}(X) = \text{Span}((X \setminus \{x_i\}) \cup \{v\}).$$

In words, we may *exchange* the vector x_i for v without changing the span.

Proof. Without loss of generality, take $i = 1$. Rearranging,

$$x_1 = \lambda_1^{-1}(v - \lambda_2 x_2 - \cdots - \lambda_n x_n),$$

so $x_1 \in \text{Span}((X \setminus \{x_1\}) \cup \{v\})$. Hence

$$\text{Span}((X \setminus \{x_1\}) \cup \{v\}) = \text{Span}(X \cup \{v\}).$$

But $v \in \text{Span}(X)$, so $\text{Span}(X \cup \{v\}) = \text{Span}(X)$. Thus

$$\text{Span}(X) = \text{Span}((X \setminus \{x_1\}) \cup \{v\}).$$

□

Independent vs. spanning: the finite case

Theorem 3.24. Let $X \subseteq V$ be linearly independent and $Y \subseteq V$ be spanning, and suppose $|Y| < \infty$. Then

$$|X| \leq |Y|.$$

Proof. First reduce to the case $|X| < \infty$. If $|X| = \infty$ and $|Y| = m$, then by definition of infinite cardinality we can choose a finite subset $X' \subseteq X$ with $|X'| = m + 1$. If we prove the theorem for finite linearly independent sets, we obtain a contradiction for X' and Y , so it suffices to consider the case where $|X|$ is finite.

We will prove in Theorem 3.25 below that there exists a spanning set $Y' \subseteq X \cup Y$ such that $X \subseteq Y'$ and $|Y'| = |Y|$. Then $|Y| = |Y'| \geq |X|$, which gives the desired inequality. □

Now assume X and Y are both finite, with

$$X = \{x_1, \dots, x_m\} \text{ linearly independent, } Y = \{y_1, \dots, y_u\} \text{ spanning.}$$

Theorem 3.25. There exists a subset $Y' \subseteq X \cup Y$ such that

$$Y' \supseteq X, \quad |Y'| = |Y|, \quad Y' \text{ is spanning.}$$

Note 3.26. As already observed, $Y' \supseteq X$ implies $|X| \leq |Y'| = |Y|$, so Theorem 3.25 implies Theorem 3.24.

Proof of Theorem 3.25. We argue by induction on $|X \setminus Y|$.

If $|X \setminus Y| = 0$, then $X \subseteq Y$. Since Y is spanning, we can simply take $Y' = Y$, which clearly satisfies the required properties.

Now suppose $|X \setminus Y| = k > 0$. Choose $x \in X \setminus Y$. Since Y is spanning, we can write

$$x = \lambda_1 y_1 + \dots + \lambda_n y_n$$

for some $y_1, \dots, y_n \in Y$ and scalars $\lambda_i \in F$, not all zero. By discarding zero coefficients if necessary, we may assume all $\lambda_i \neq 0$.

We claim that at least one of the y_i does not lie in X . If all y_i belonged to X , then we would have a non-trivial linear relation

$$-x + \lambda_1 y_1 + \dots + \lambda_n y_n = 0$$

among distinct elements of X (note that $x \notin Y$ so $x \neq y_i$ for all i), with all coefficients non-zero. This contradicts linear independence of X . Hence some $y_i \notin X$.

Define

$$Y' := (Y \setminus \{y_i\}) \cup \{x\}.$$

By the Steinitz Exchange Lemma, $\text{Span}(Y') = \text{Span}(Y)$, so Y' is also spanning. Moreover,

$$X \setminus Y' = (X \setminus Y) \setminus \{x\},$$

so $|X \setminus Y'| = |X \setminus Y| - 1 = k - 1$. We also have $X \cup Y' \subseteq X \cup Y$ and $|Y'| = |Y|$.

By the induction hypothesis applied to X and Y' , there exists a spanning set $Y'' \subseteq X \cup Y'$ such that $X \subseteq Y''$ and $|Y''| = |Y'| = |Y|$. Since $X \cup Y' \subseteq X \cup Y$, this Y'' also lies in $X \cup Y$, and the proof is complete. □

Finite-dimensional vector spaces and bases

Definition 3.27. A vector space V is called *finite-dimensional* if it has a finite spanning set $X \subseteq V$.

Corollary 3.28. Any two bases of a finite-dimensional vector space V have the same cardinality.

Proof. Let B_1 and B_2 be bases of V , and let $X \subseteq V$ be a finite spanning set. By Theorem 3.24,

$$|B_1| \leq |X|, \quad |B_2| \leq |X| < \infty.$$

Applying the same theorem with $X = B_1$ and $Y = B_2$ gives $|B_1| \leq |B_2|$, and with $X = B_2$, $Y = B_1$ gives $|B_2| \leq |B_1|$. Hence $|B_1| = |B_2|$. $\square$

Proposition 3.29. Suppose $X \subseteq V$ is a *minimal* spanning set, meaning that for every proper subset $X' \subsetneq X$, the set X' is not spanning. Then X is a basis of V .

Proof. We already know X is spanning. Suppose, for contradiction, that X is linearly dependent. Then there exists a non-trivial relation

$$\lambda_1 x_1 + \cdots + \lambda_n x_n = 0,$$

with all $\lambda_i \neq 0$, among distinct elements $x_1, \dots, x_n \in X$. By the Steinitz Exchange Lemma, we can replace one of the x_i by 0 and still obtain a spanning set; in particular,

$$(X \setminus \{x_i\}) \cup \{0\}$$

spans V , and hence $X \setminus \{x_i\}$ is spanning. This contradicts the minimality of X . Therefore X must be linearly independent, and hence is a basis. $\square$

Corollary 3.30. If $Y \subseteq V$ is a finite spanning set, then there exists a subset $X \subseteq Y$ which is a basis of V .

Proof. Let Y be spanning with $|Y| = n < \infty$. If Y is linearly independent, we are done. Otherwise, Y is linearly dependent, so there exists $v \in Y$ such that $Y \setminus \{v\}$ is still spanning (by the previous lemma (ii)). If this new set is still linearly dependent, we repeat the process and remove another redundant vector. Since Y is finite, this procedure terminates, and we obtain a minimal spanning subset $X \subseteq Y$, which must be a basis by the previous proposition.

This procedure is constructive: it gives an algorithm to extract a basis from a finite spanning set. $\square$

Definition 3.31. If V is finite-dimensional, the *dimension* of V , denoted $\dim V$, is defined to be $|X|$ for any basis X of V . This is well-defined by the corollary above.

Proposition 3.32. If $X \subseteq V$ is a *maximal* linearly independent set (meaning that X is linearly independent and no proper superset of X is linearly independent), then X is a basis of V .

Proof. If $v \notin \text{Span}(X)$, then by Lemma 3.21.(i), the set $X \cup \{v\}$ is linearly independent, contradicting maximality. Hence every vector in V lies in $\text{Span}(X)$, so X is spanning and therefore a basis. $\square$

Corollary 3.33. Let V be finite-dimensional, and let $X \subseteq V$ be linearly independent and $Y \subseteq V$ be spanning. Then there exists a basis B of V such that

$$X \subseteq B \subseteq X \cup Y.$$

Proof. Let $n = \dim V < \infty$ and $|X| = m \leq n$. If X is already a basis, set $B = X$. Otherwise, $\text{Span}(X) \subsetneq V = \text{Span}(Y)$, so $\text{Span}(Y) \not\subseteq \text{Span}(X)$. Applying the contrapositive of Corollary 3.12, we deduce that $Y \not\subseteq \text{Span}(X)$. Hence there exists $y_1 \in Y$ with $y_1 \notin \text{Span}(X)$, and $X_1 := X \cup \{y_1\}$ is linearly independent.

If X_1 is not a basis, we repeat: $\text{Span}(X_1) \subsetneq V$, so there exists $y_2 \in Y$ with $y_2 \notin \text{Span}(X_1)$, and we set $X_2 := X_1 \cup \{y_2\}$. Continuing in this way, we obtain a strictly increasing chain of linearly independent sets

$$X = X_0 \subsetneq X_1 \subsetneq X_2 \subsetneq \cdots$$

with $|X_k| = m + k$ and $X_k \subseteq X \cup Y$. Since V is finite-dimensional, we cannot continue indefinitely: once $|X_k| = n$, the set X_k must be a basis. Let B be the first such X_k that spans V ; then B is a basis containing X and contained in $X \cup Y$. $\square$

Corollary 3.34. Let V be finite-dimensional with $n := \dim V$. The following are equivalent for a subset $X \subseteq V$:

1. X is a basis of V .
2. X is linearly independent and $|X| = n$.
3. X is spanning and $|X| = n$.

In particular, if $|X| = n$, then

$$X \text{ is linearly independent} \iff X \text{ is spanning} \iff X \text{ is a basis.}$$

Proof. Clearly (1) implies both (2) and (3).

(2) $\Rightarrow$ (1): Let X be linearly independent with $|X| = n$. If X were not a basis, then we could add a vector $v \notin \text{Span}(X)$ and obtain a linearly independent set of size $n + 1$, contradicting the fact that $\dim V = n$.

(3) $\Rightarrow$ (1): Let X be spanning with $|X| = n$. If X were not linearly independent, then there would exist a proper subset $X' \subsetneq X$ which is still spanning, with $|X'| < n$, contradicting $\dim V = n$. $\square$

Examples 3.35. • $\dim F^n = n$, with basis $e_1, \dots, e_n$.

- $\dim M_{m,n}(F) = mn$. A natural basis is given by the matrices $E^{(i,j)}$ with a single 1 in position (i, j) and zeros elsewhere:

$$(E^{(i,j)})_{k\ell} = \begin{cases} 1, & \text{if } k = i, \ell = j, \\ 0, & \text{otherwise.} \end{cases}$$

Dimension of subspaces

Proposition 3.36. If V is finite-dimensional and $U \subseteq V$ is a subspace, then U is also finite-dimensional and

$$\dim U \leq \dim V.$$

Moreover, $\dim U = \dim V$ if and only if $U = V$.

Proof. Any basis of U is a linearly independent subset of V , so its cardinality cannot exceed $\dim V$, giving $\dim U \leq \dim V$.

We must also show that such a basis exists. Start with the empty set, which is linearly independent in U , and repeatedly adjoin vectors of U that are not yet in the span. Since no linearly independent subset of V can have more than $\dim V$ elements, this process terminates in at most $\dim V$ steps and yields a finite basis of U .

If $\dim U = \dim V$, then a basis of U is a linearly independent subset of V of size $\dim V$, hence it is a basis of V by the previous corollary, so $U = V$. Conversely, if $U = V$ then clearly $\dim U = \dim V$. $\square$

Examples

- For $V = F^1 = F$, the only subspaces are $\{0\}$ and F .
- For $V = F^2$, the subspaces are:
 - $\{0\}$ (dimension 0);
 - lines through the origin of the form $\{(\lambda a, \lambda b) : \lambda \in F\}$ with $(a, b) \neq (0, 0)$ (dimension 1);
 - the whole space F^2 (dimension 2).
- For $V = F^3$, the subspaces are:
 - $\{0\}$;
 - lines through the origin: $\{(\lambda a, \lambda b, \lambda c) : \lambda \in F\}$, $(a, b, c) \neq (0, 0, 0)$;
 - planes through the origin: $\{(x, y, z) : ax + by + cz = 0\}$ with $(a, b, c) \neq (0, 0, 0)$;
 - the whole space F^3 .

General observations

- 0-dimensional subspaces are exactly $\{0\}$.
- 1-dimensional subspaces of V are of the form

$$F \cdot v = \{\lambda v : \lambda \in F\}, \quad v \in V \setminus \{0\}.$$

- 2-dimensional subspaces are of the form $\text{Span}(v_1, v_2)$ with $v_1, v_2 \neq 0$ and $v_2 \notin F \cdot v_1$.

Uniqueness of expansions in a basis

Proposition 3.37. Let V be a vector space and $X, Y, B \subseteq V$.

- (i) A subset X is linearly independent if and only if every vector $v \in V$ has *at most one* representation

$$v = \lambda_1 x_1 + \cdots + \lambda_m x_m,$$

as a linear combination of distinct elements $x_1, \dots, x_m \in X$ with $\lambda_i \neq 0$.

- (ii) A subset Y is spanning if and only if every $v \in V$ has *at least one* such representation with $x_i \in Y$.

- (iii) A subset B is a basis if and only if every $v \in V$ has *exactly one* expression as a linear combination of distinct elements of B with nonzero coefficients.

Proof. (i) If X is linearly independent and v had two expressions

$$v = \lambda_1 x_1 + \cdots + \lambda_m x_m = \lambda'_1 x'_1 + \cdots + \lambda'_n x'_n$$

with $x_1, \dots, x_m \in X$ distinct and $x'_1, \dots, x'_n \in X$ distinct, and λ_i, λ'_i nonzero, then let us reorder the variables such that $x_1 = x'_1, \dots, x_k = x'_k$, and $x_1, \dots, x_m, x'_{k+1}, \dots, x'_n$ are all distinct.

Then subtracting the two expressions gives:

$$(\lambda_1 - \lambda'_1)x_1 + \cdots + (\lambda_k - \lambda'_k)x_k + \lambda_{k+1}x_{k+1} + \cdots + \lambda_m x_m - \lambda'_{k+1}x'_{k+1} - \cdots - \lambda'_n x'_n = 0.$$

Linear independence forces $\lambda_i = \lambda'_i$ for all $1 \leq i \leq k$, $\lambda_i = 0$ for $i > k$, and $\lambda'_i = 0$ for $i > k$. This implies by assumption that $m = n = k$ and the two expressions were the same. Thus there can be only one expression.

Conversely, if X admits a non-trivial dependence, then 0 has at least two distinct representations (the zero combination and a non-trivial one), contradicting uniqueness.

(ii) This is simply the definition of $\text{Span}(Y)$.

(iii) A basis is by definition linearly independent and spanning, so combining (i) and (ii) gives both existence and uniqueness of expressions (for existence, we hadn't explicitly assumed that the x_i were unique and the coefficients nonzero, but one can simply combine terms and discard terms with coefficient equal to zero, so that this is true). Conversely, if every vector has a unique expression in terms of B , then both existence and uniqueness hold, so B is simultaneously spanning and linearly independent, hence a basis. $\square$

Remark 3.38. It is common to *define* a basis as a subset $B \subseteq V$ such that each $v \in V$ has a unique expression as a finite linear combination of elements of B .

The dimension formula for sums of subspaces

Theorem 3.39 (Thm 3.7.9). Let $U, W \subseteq V$ be subspaces, and set

$$U + W := \{u + w : u \in U, w \in W\}.$$

Then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. Since V is finite-dimensional, so are U , W , and $U \cap W$. Let

$$(v_1, \dots, v_m)$$

be a basis of $U \cap W$. Extend this basis to a basis of U by adding vectors $u_1, \dots, u_n$:

$$(v_1, \dots, v_m, u_1, \dots, u_n),$$

and to a basis of W by adding vectors $w_1, \dots, w_p$:

$$(v_1, \dots, v_m, w_1, \dots, w_p).$$

We will show that the list

$$(v_1, \dots, v_m, u_1, \dots, u_n, w_1, \dots, w_p)$$

is a basis of $U + W$; this yields the formula for the dimension for the following reason: in this case

$$\dim(U + W) = m + n + p.$$

But

$$\dim U = m + n, \quad \dim W = m + p, \quad \dim(U \cap W) = m,$$

so

$$\dim(U + W) = (m + n + p) = (m + n) + (m + p) - m = \dim U + \dim W - \dim(U \cap W). \quad \square$$

Proposition 3.40. The list

$$(v_1, \dots, v_m, u_1, \dots, u_n, w_1, \dots, w_p)$$

constructed above is a basis of $U + W$.

Proof. Linear independence. Suppose

$$\lambda_1 v_1 + \dots + \lambda_m v_m + \mu_1 u_1 + \dots + \mu_n u_n + \nu_1 w_1 + \dots + \nu_p w_p = 0.$$

Rewrite the equation as:

$$\lambda_1 v_1 + \dots + \lambda_m v_m + \mu_1 u_1 + \dots + \mu_n u_n = -\nu_1 w_1 - \nu_2 w_2 - \dots - \nu_p w_p.$$

Call the vector on both sides of this equation v . The left-hand side belongs to U , since $v_i, u_j \in U$, and right-hand side belongs to W , since $w_j \in W$. Hence $v \in U \cap W$. Therefore it is a linear combination of $v_1, \dots, v_m$, say $v = \lambda'_1 v_1 + \dots + \lambda'_m v'_m$. We therefore get two expressions for v in the basis $(v_1, \dots, v_m, u_1, \dots, u_n)$ of U :

$$v = \lambda'_1 v_1 + \dots + \lambda'_m v_m + 0u_1 + \dots + 0u_n = \lambda_1 v_1 + \dots + \lambda_m v_m + \mu_1 u_1 + \dots + \mu_n u_n.$$

By Proposition 3.37.(i), we must have $\lambda'_i = \lambda_i$ for all i , and $\mu_i = 0$ for all i .

Next, we have two expressions for v in the basis $(v_1, \dots, v_m, w_1, \dots, w_p)$ of W :

$$v = \lambda_1 v_1 + \dots + \lambda_m v_m + 0w_1 + \dots + 0w_p = 0v_1 + \dots + 0v_m + \nu_1 w_1 + \dots + \nu_p w_p.$$

Therefore we get, by Proposition 3.37.(i), that $\lambda_i = 0$ for all i and $\nu_i = 0$ for all i . This shows the linear independence of $(v_1, \dots, v_m, u_1, \dots, u_n, w_1, \dots, w_p)$.

Spanning. Any vector $u \in U$ can be written as a linear combination of $(v_1, \dots, v_m, u_1, \dots, u_n)$, and any $w \in W$ as a linear combination of $(v_1, \dots, v_m, w_1, \dots, w_p)$. Thus any vector in $U + W$ is a sum of such combinations and hence lies in the span of $(v_1, \dots, v_m, u_1, \dots, u_n, w_1, \dots, w_p)$. This shows that the list spans $U + W$. $\square$

Remark 3.41. The argument used in spanning (which is the easy part) generalises to the following: For subsets $X, Y \subseteq V$,

$$\text{Span}(X) + \text{Span}(Y) = \text{Span}(X \cup Y).$$

To prove this, note that the inclusion $\subseteq$ holds because any sum of a vector in $\text{Span}(X)$ and a vector in $\text{Span}(Y)$ is a linear combination of elements of $X \cup Y$.

For the reverse inclusion, note that every element of $\text{Span}(X \cup Y)$ can be written as a linear combination of elements of X and of Y . By regrouping terms, this is a sum of an element of $\text{Span}(X)$ and an element of $\text{Span}(Y)$. Thus we get the inclusion $\supseteq$.

Example 3.42. If $\dim U = \dim W = 1$, say $U = F \cdot u$ and $W = F \cdot w$, then:

- If $U = W$, then $U = W = U \cap W = U + W$ and $\dim(U + W) = 1 = 1 + 1 - 1$.
- If $U \neq W$, then $U \cap W = \{0\}$ and $U + W$ is a 2-dimensional subspace, so $\dim(U + W) = 2 = 1 + 1 - 0$.

Example 3.43. Let $V = F^n$, and consider the subspaces

$$U = \{(x_1, \dots, x_m, 0, \dots, 0)\}, \quad W = \{(0, \dots, 0, y_1, \dots, y_p)\},$$

where the x_i occupy the first m coordinates and the y_j the last p coordinates.

If $m + p \leq n$, then $U \cap W = \{0\}$ and

$$U + W \cong F^{m+p},$$

so $\dim(U \cap W) = 0$ and $\dim(U + W) = m + p = \dim U + \dim W - 0$.

If $m + p \geq n$, then

$$U \cap W \cong F^{m+p-n},$$

so $\dim(U \cap W) = m + p - n$, whilst $U + W = V \cong F^n$, so $\dim(U + W) = n = m + p - (m + p - n)$, in accordance with the formula.

Row space, column space, and rank

Proposition 3.44. Let $A \in M_{m \times n}(F)$. Elementary row operations do not change the row space of A .

Proof. If A' is obtained from A by a sequence of row operations, then each row of A' is a linear combination of rows of A , so

$$\text{rowspace}(A') \subseteq \text{rowspace}(A).$$

Conversely, since row operations are invertible, each row of A is a linear combination of rows of A' . Hence

$$\text{rowspace}(A) \subseteq \text{rowspace}(A'),$$

and the two row spaces coincide. $\square$

Corollary 3.45. The dimension of the row space of A is equal to the number of non-zero rows (equivalently, the number of pivot columns) in a row echelon form of A .

Proof. In row echelon form, the non-zero rows have their first non-zero entries in distinct columns. Any linear relation among the non-zero rows

$$\lambda_1 v_1 + \dots + \lambda_k v_k = 0$$

must have all $\lambda_i = 0$, which can be seen by inspecting the pivot positions from top to bottom. Thus the non-zero rows form a linearly independent spanning set for the row space, and their number is the dimension of the row space. This number is the same as the number of pivots. $\square$

Proposition 3.46. Let $A \in M_{m \times n}(F)$ with columns $v_1, \dots, v_n$. The solutions to

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0$$

are unaffected by row operations on A .

Proof. The equation can be written in matrix form as

$$A \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = 0.$$

If A' is obtained from A by an elementary row operation, then for $\lambda \in F^n$ as above,

$$A'\lambda = 0 \iff A\lambda = 0.$$

So the set of solutions is the same for A and A' . □

Corollary 3.47. Elementary row operations do not change which subsets of columns of A are linearly independent.

Corollary 3.48. If the pivot columns of a row echelon form of A are columns $i_1, \dots, i_k$, then

$$\{v_{i_1}, \dots, v_{i_k}\}$$

is a basis of the column space of A .

Proof. In reduced row echelon form, the pivot columns are standard basis vectors, which form a maximal linearly independent set; their pre-images in the original matrix are therefore linearly independent and span the same space, so they form a basis of the column space. □

Corollary 3.49. For any matrix A ,

$$\dim \text{rowspace}(A) = \dim \text{colspace}(A).$$

This common dimension is equal to the number of pivots in a row echelon form of A .

Definition 3.50. The common dimension of the row space and the column space of A is called the *rank* of A and is denoted $\text{rk}(A)$.

Remark 3.51. One can also prove $\dim \text{rowspace}(A) = \dim \text{colspace}(A)$ without using row operations, via factorizations $A = BC$ with $B \in M_{m \times k}(F)$ and $C \in M_{k \times n}(F)$.

Example 3.52. Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 6 & 6 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{pmatrix}.$$

If $3 \neq 0$ in F , we can further reduce to

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The non-zero rows span the row space, so

$$\text{rowspace}(A) = \text{Span}((1, 2, 3), (0, 3, 6)).$$

The pivot columns are the first and second, so

$$\text{colspace}(A) = \text{Span} \left(\begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} \right).$$

These are bases provided $3 \neq 0$ in F . In this case,

$$\text{rk}(A) = 2;$$

if $3 = 0$ in F , then $\text{rk}(A) = 1$.

Thus the rank of A can be computed as the number of pivots. This is also the dimension of $\text{Span}(v_1, \dots, v_m)$ if we treat the v_i either as row vectors or as column vectors.

The rank-nullity theorem for matrices

Definition 3.53. The nullspace, or kernel, of a matrix A is the set of vectors v such that $Av = 0$.

In other words, this is the set of solutions of the homogeneous equations with coefficients in A .

Theorem 3.54. We have the equality for $A \in M_{m,n}(F)$:

$$\text{rk}(A) + \dim \text{nullspace}(A) = n.$$

Proof. We saw that the nullspace has number of parameters equal to the number of free variables. In fact, this shows that a basis for the nullspace is of this size: if $R = \text{rref } A$, and the free variables are $x_{i_1}, \dots, x_{i_k}$, then the solution set is spanned by the unique solutions $v_1, \dots, v_k$ to $Av = 0$ such that the i_j -th entry of v_j is 1 and the i_ℓ -th entry is 0 for all $\ell \neq j$. Clearly, the $v_1, \dots, v_k$ are linearly independent, since $\lambda_1 v_1 + \dots + \lambda_k v_k$ has, in the free entries, $\lambda_1, \dots, \lambda_k$, so if this is zero, all the λ_i are zero. Thus, it is a basis, as desired.

Then, the formula follows from the one: #pivot variables + # free variables = n . $\square$

Right and left invertibility of matrices, and rank

Proposition 3.55. The following are equivalent for a square matrix $A \in M_{n,n}(F)$:

1. A has a right inverse
2. A has a left inverse
3. $\text{rk}(A) = n$
4. A is invertible

Proof. We prove that (a) is equivalent to (c). Suppose that $B = (b_{ij})$. Write the columns of A as $v_1, \dots, v_n$ and the columns of AB as $w_1, \dots, w_n$. Then we have:

$$w_j = \sum_i b_{ij} v_i.$$

So solving $AB = I$ is the same as finding, in the column space of A , all the vectors $e_1, \dots, e_n$. This is possible if and only if $\text{colspace}(A) = n$. By definition of rank, we conclude the statement.

By transposing, we get that (b) $\Leftrightarrow$ (c) as well. So (a), (b), and (c) are all equivalent. Since also (a) and (b) together are equivalent to (d), we conclude the statement. $\square$

Remark 3.56. In Corollary 2.52, we already showed that (a) is equivalent to $\text{rref}(A) = I$, which by our definition of rank, is equivalent to (c). So we can alternatively deduce the proof from that statement.

Gaussian elimination and bases of subspaces

Let $V = \text{Span}(v_1, \dots, v_m) \subseteq F^n$.

There are two convenient ways to use Gaussian or Gauss–Jordan elimination to find a basis of V :

1. View F^n as row vectors, and form the matrix

$$A = \begin{pmatrix} v_1 \\ \vdots \\ v_m \end{pmatrix}.$$

Perform Gaussian or Gauss–Jordan elimination to obtain a row echelon (or reduced row echelon) matrix R . Then

$$V = \text{rowspace}(A) = \text{rowspace}(R),$$

and a basis of V is given by the non-zero rows of R .

2. View F^n as column vectors, and form the matrix

$$A = (v_1 \ \cdots \ v_m).$$

Perform Gaussian or Gauss–Jordan elimination to obtain a row echelon matrix R . If the pivot columns of R are $i_1, \dots, i_k$, then

$$(v_{i_1}, \dots, v_{i_k})$$

is a basis of the column space of A , that is, of $V = \text{Span}(v_1, \dots, v_m)$. In this way we obtain a basis as a subset of the original generating set.

Bonus: determinants

Note: this topic will not be assumed knowledge for the January exam.

Proposition 3.57. There exists a function

$$\det : M_{n \times n}(F) \longrightarrow F$$

such that:

- If A' is obtained from A by swapping two rows, then $\det A' = -\det A$.
- If A' is obtained from A by multiplying one row by a scalar $\lambda \in F$, then $\det A' = \lambda \det A$.
- If A' is obtained from A by adding a multiple of one row to another, then $\det A' = \det A$.
- $\det I_n = 1$.

Corollary 3.58. Let A be reduced to $R = \text{rref}(A)$ via Gauss–Jordan elimination. Then

$$\det A = \begin{cases} 0, & \text{if } R \neq I_n, \\ (-1)^k \lambda_1 \cdots \lambda_m, & \text{if } R = I_n, \end{cases}$$

where k is the number of row swaps performed and $\lambda_1, \dots, \lambda_m$ are the scalars used in row rescalings.

Corollary 3.59. A square matrix A is invertible if and only if $\det A \neq 0$.

Exercise 3.60. Show that

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

and

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & i \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}.$$

Fact. For any $A \in M_{n \times n}(F)$,

$$\det A = a_{11} \det A^{11} - a_{12} \det A^{12} + \cdots + (-1)^{1+n} a_{1n} \det A^{1n},$$

where A^{ij} denotes the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j (the *cofactor expansion* along the first row).

4 Linear Transformations

This is really what linear algebra is about: linear maps between vector spaces.

Definition 4.1. Let $T : V \rightarrow W$ be a map between vector spaces over a field F . We say that T is *linear* if, for all $u, v \in V$ and $\lambda \in F$,

$$\begin{aligned} T(u + v) &= T(u) + T(v), \\ T(\lambda v) &= \lambda T(v). \end{aligned}$$

- A linear map $T : V \rightarrow W$ is a (linear) *isomorphism* if it is bijective.
- If T is an isomorphism, then its inverse $T^{-1} : W \rightarrow V$ is also linear (see PS4, B4).

Connections to matrices

Proposition 4.2. Let $T : F^n \rightarrow F^m$ be a map. Then

$$T \text{ is linear} \iff \exists A \in M_{m \times n}(F) \text{ such that } T(v) = Av$$

for all $v \in F^n$, where we regard $F^n \cong M_{n,1}(F)$ and $F^m \cong M_{m,1}(F)$ as column vectors. Moreover, the matrix A is unique.

Proof. Assume T is linear. Let $e_1, \dots, e_n$ be the standard basis of F^n . For

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \lambda_1 e_1 + \cdots + \lambda_n e_n,$$

we have

$$T(v) = T(\lambda_1 e_1 + \cdots + \lambda_n e_n) = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n).$$

Define A to be the matrix whose i -th column is $T(e_i)$:

$$A = (T(e_1) \cdots T(e_n)).$$

Then for every $v \in F^n$ we have $T(v) = Av$.

Conversely, if $T(v) = Av$ for some fixed matrix A , then T is linear because matrix multiplication is linear in v . Uniqueness of A follows from the fact that a linear map is determined by its values on a basis: the columns $T(e_i)$ are uniquely determined, hence so is A . $\square$

Matrices with respect to general bases

Let V and W be finite-dimensional vector spaces over F . Fix a basis

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{of } V, \quad \mathcal{C} = (w_1, \dots, w_m) \quad \text{of } W.$$

For any $v \in V$ there is a unique expression

$$v = \lambda_1 v_1 + \dots + \lambda_n v_n,$$

and we write its *coordinate column vector* in the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} := \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}.$$

Similarly, for $w \in W$ we define

$$[w]_{\mathcal{C}} := \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_m \end{pmatrix} \quad \text{if } w = \mu_1 w_1 + \dots + \mu_m w_m.$$

Proposition 4.3. Let $T : V \rightarrow W$ be linear, and let $\mathcal{B}, \mathcal{C}$ be bases as above. Then there exists a unique matrix

$${}_c T_{\mathcal{B}} \in M_{m \times n}(F)$$

such that, for every $v \in V$,

$$[T(v)]_{\mathcal{C}} = {}_c T_{\mathcal{B}} [v]_{\mathcal{B}}.$$

Equivalently, if $T(v_i) = \sum_{j=1}^m A_{ji} w_j$, then

$${}_c T_{\mathcal{B}} = (A_{ji})_{1 \leq j \leq m, 1 \leq i \leq n}$$

is the matrix whose i -th column is $[T(v_i)]_{\mathcal{C}}$.

Definition 4.4. The matrix ${}_c [T]_{\mathcal{B}}$ is called the *matrix of T with respect to the bases $\mathcal{B}$ (in V) and $\mathcal{C}$ (in W)*.

Proof of Proposition 4.3. For $v \in V$ we have a unique expression $v = \lambda_1 v_1 + \dots + \lambda_n v_n$ and thus a unique coordinate vector $[v]_{\mathcal{B}}$. Likewise, $T(v)$ has a unique expression

$$T(v) = \mu_1 w_1 + \dots + \mu_m w_m$$

and hence a unique coordinate vector $[T(v)]_{\mathcal{C}}$.

Define the map

$$\Phi : F^n \rightarrow F^m, \quad \Phi([v]_{\mathcal{B}}) := [T(v)]_{\mathcal{C}}.$$

By linearity of T , the map Φ is linear:

$$\begin{aligned} [v]_{\mathcal{B}} &= \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}, \quad [v']_{\mathcal{B}} = \begin{pmatrix} \lambda'_1 \\ \vdots \\ \lambda'_n \end{pmatrix}, \\ [v + v']_{\mathcal{B}} &= \begin{pmatrix} \lambda_1 + \lambda'_1 \\ \vdots \\ \lambda_n + \lambda'_n \end{pmatrix}, \\ [T(v + v')]_{\mathcal{C}} &= [T(v)]_{\mathcal{C}} + [T(v')]_{\mathcal{C}}, \end{aligned}$$

and similarly for scalar multiples. By Proposition 4.2, there is a unique matrix $A = {}_c[T]_{\mathcal{B}}$ such that

$$[T(v)]_{\mathcal{C}} = {}_c[T]_{\mathcal{B}} [v]_{\mathcal{B}}$$

for all v , or in other words, such that $\Phi = T_A$.

Finally, to compute the matrix A , note that, if the columns of A are $A_1, \dots, A_n$, then $A_i = Ae_i = \Phi(e_i) = [Tv_i]_{\mathcal{C}}$. $\square$

Conceptual picture via coordinate isomorphisms

A basis $\mathcal{B} = (v_1, \dots, v_n)$ of V gives an isomorphism

$$[\cdot]_{\mathcal{B}} : V \xrightarrow{\sim} F^n, \quad v = \lambda_1 v_1 + \dots + \lambda_n v_n \mapsto [v]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}.$$

Similarly, a basis $\mathcal{C}$ of W gives $[\cdot]_{\mathcal{C}} : W \xrightarrow{\sim} F^m$.

For a linear map $T : V \rightarrow W$ we then have a commuting square, for $A = {}_c[T]_{\mathcal{B}}$:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \downarrow [\cdot]_{\mathcal{B}} & & \downarrow [\cdot]_{\mathcal{C}} \\ F^n & \xrightarrow{T_A} & F^m. \end{array}$$

Examples in $\mathbb{R}^2$

Let $V = W = \mathbb{R}^2$ with the standard basis $e_1 = (1, 0)$, $e_2 = (0, 1)$.

- Rotation by angle θ :

$$R_{\theta} : \mathbb{R}^2 \rightarrow \mathbb{R}^2.$$

We have

$$R_{\theta}(e_1) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad R_{\theta}(e_2) = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}.$$

Hence, with respect to the standard basis $\mathcal{E} = (e_1, e_2)$,

$${}_{\mathcal{E}}[R_{\theta}]_{\mathcal{E}} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Thus

$$R_{\theta} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

- Reflection across a line making angle θ with the x -axis:

$$\text{Ref}_{\theta} \begin{pmatrix} a \\ b \end{pmatrix} = A \begin{pmatrix} a \\ b \end{pmatrix}, \quad A = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

So ${}_{\mathcal{E}}[\text{Ref}_{\theta}]_{\mathcal{E}} = A$.

Examples in polynomial spaces

Let $V = F[x]_{\leq m}$ and $W = F[x]_{\leq m-1}$ be the spaces of polynomials of degree at most m and at most $m-1$ respectively. Consider

$$T = \frac{d}{dx} : V \rightarrow W.$$

With respect to the standard monomial bases

$$\mathcal{B} = (1, x, \dots, x^m) \quad \text{of } V, \quad \mathcal{C} = (1, x, \dots, x^{m-1}) \quad \text{of } W,$$

we have

$$\frac{d}{dx}(x^k) = kx^{k-1}.$$

Thus the matrix of T in these bases is

$$c[T]\mathcal{B} = {}_{(1,x,\dots,x^{m-1})} \left[\left(\frac{d}{dx} \right) \right]_{(1,x,\dots,x^m)} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & m \end{pmatrix}.$$

Now consider an “integration” map (ignoring constants of integration for the moment)

$$S : W \rightarrow V, \quad S(x^k) = \frac{x^{k+1}}{k+1},$$

and then allow constants by writing

$$\int x^k dx = \frac{x^{k+1}}{k+1} + c.$$

With respect to the bases

$$\mathcal{C} = (1, x, \dots, x^{m-1}), \quad \mathcal{B} = (1, x, \dots, x^m),$$

the matrix

$$\mathcal{B}[S]\mathcal{C} = {}_{(1,x,\dots,x^m)} \left[\left(\int dx \right) \right]_{(1,x,\dots,x^{m-1})}$$

has the form

$$\mathcal{B}[S]\mathcal{C} = \begin{pmatrix} c_0 & c_1 & \cdots & c_{m-1} \\ 1 & 0 & \cdots & 0 \\ 0 & \frac{1}{2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{m} \end{pmatrix},$$

for some constants $c_0, \dots, c_{m-1}$ representing the choice of integration constants.

One checks that

$$c[T]\mathcal{B}\mathcal{B}[S]\mathcal{C} = I_m,$$

so (up to the choice of constants) differentiation followed by integration gives back the original polynomial in W .

Composition of linear maps

Let

$$U \xrightarrow{S} V \xrightarrow{T} W$$

be linear maps, with bases

$$\mathcal{B} = (u_1, \dots, u_m) \text{ for } U, \quad \mathcal{C} = (v_1, \dots, v_n) \text{ for } V, \quad \mathcal{D} = (w_1, \dots, w_p) \text{ for } W.$$

Proposition 4.5. With the notation above,

$$\mathcal{D}[T \circ S]_{\mathcal{B}} = \mathcal{D}[T]_{\mathcal{C}} \cdot \mathcal{C}[S]_{\mathcal{B}}.$$

Proof. For $u \in U$ write $[u]_{\mathcal{B}} = (\lambda_1, \dots, \lambda_m)^{\top}$. Then

$$[S(u)]_{\mathcal{C}} = \mathcal{C}[S]_{\mathcal{B}} [u]_{\mathcal{B}},$$

and

$$[T(S(u))]_{\mathcal{D}} = \mathcal{D}[T]_{\mathcal{C}} [S(u)]_{\mathcal{C}} = \mathcal{D}[T]_{\mathcal{C}} \mathcal{C}[S]_{\mathcal{B}} [u]_{\mathcal{B}}.$$

On the other hand, by definition,

$$[T \circ S(u)]_{\mathcal{D}} = \mathcal{D}[T \circ S]_{\mathcal{B}} [u]_{\mathcal{B}}.$$

Since this holds for all $[u]_{\mathcal{B}}$, we deduce

$$\mathcal{D}[T \circ S]_{\mathcal{B}} = \mathcal{D}[T]_{\mathcal{C}} \cdot \mathcal{C}[S]_{\mathcal{B}}.$$

□

Alternative conceptual proof. Using the coordinate isomorphisms, we can write

$$\mathcal{D}[T \circ S]_{\mathcal{B}} = [\cdot]_{\mathcal{D}} \circ [T \circ S] \circ [\cdot]_{\mathcal{B}}^{-1} = ([\cdot]_{\mathcal{D}} \circ [T] \circ [\cdot]_{\mathcal{C}}^{-1}) \circ ([\cdot]_{\mathcal{C}} \circ [S] \circ [\cdot]_{\mathcal{B}}^{-1}) = \mathcal{D}[T]_{\mathcal{C}} \circ \mathcal{C}[S]_{\mathcal{B}}.$$

Change-of-basis formula

Let $\mathcal{B}, \mathcal{B}'$ be two bases of V , and $\mathcal{C}, \mathcal{C}'$ two bases of W . For the identity map

$$I_V : V \rightarrow V, \quad I_W : W \rightarrow W,$$

we define the *change-of-basis matrices*

$$\mathcal{B}[I]_{\mathcal{B}'}, \quad \mathcal{C}[I]_{\mathcal{C}}.$$

These satisfy, by definition:

$$[v]_{\mathcal{B}} = \mathcal{B}[I]_{\mathcal{B}'} [v]_{\mathcal{B}'}, \quad [w]_{\mathcal{C}} = \mathcal{C}[I]_{\mathcal{C}'} [w]_{\mathcal{C}'}.$$

Corollary 4.6 (Change-of-basis for matrices of T). Let $T : V \rightarrow W$ be linear. Then

$$\mathcal{C}'[T]_{\mathcal{B}'} = \mathcal{C}'[I]_{\mathcal{C}} \cdot \mathcal{C}[T]_{\mathcal{B}} \cdot \mathcal{B}[I]_{\mathcal{B}'}.$$

Proof. For $v \in V$ we have

$$[T(v)]_{\mathcal{C}'} = \mathcal{C}'[I]_{\mathcal{C}} [T(v)]_{\mathcal{C}} = \mathcal{C}'[I]_{\mathcal{C}} \mathcal{C}[T]_{\mathcal{B}} [v]_{\mathcal{B}}.$$

Also,

$$[v]_{\mathcal{B}} = {}_{\mathcal{B}}[I]_{\mathcal{B}'} [v]_{\mathcal{B}'},$$

so

$$[T(v)]_{\mathcal{C}'} = \mathcal{C}'[I]_{\mathcal{C}} \mathcal{C}[T]_{\mathcal{B}} {}_{\mathcal{B}}[I]_{\mathcal{B}'} [v]_{\mathcal{B}'}.$$

By definition, $[T(v)]_{\mathcal{C}'} = \mathcal{C}'[T]_{\mathcal{B}'} [v]_{\mathcal{B}'}$, and since this holds for all $[v]_{\mathcal{B}'}$, we obtain the claimed identity. $\square$

Definition 4.7. The matrix ${}_{\mathcal{B}}[I]_{\mathcal{B}'}$ is called the *change-of-basis matrix writing the source basis $\mathcal{B}'$ in terms of the target basis $\mathcal{B}$* . If $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (v'_1, \dots, v'_n)$ and we write

$$v'_i = a_{1i}v_1 + \dots + a_{ni}v_n,$$

then

$${}_{\mathcal{B}}[I]_{\mathcal{B}'} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

whose i -th column is $[v'_i]_{\mathcal{B}}$.

This is also sometimes (in the literature) called the change-of-basis *from $\mathcal{B}$ to $\mathcal{B}'$* because it is usually the most convenient matrix to compute relating a new basis $\mathcal{B}'$ to one $\mathcal{B}$ we already had.

Corollary 4.8. For any basis $\mathcal{B}$, the matrix of the identity in that basis is the identity matrix:

$${}_{\mathcal{B}}[I]_{\mathcal{B}} = I.$$

Moreover, if we set

$$P = {}_{\mathcal{B}}[I]_{\mathcal{B}'},$$

then P is invertible and

$$P^{-1} = {}_{\mathcal{B}'}[I]_{\mathcal{B}}.$$

That is, the inverse change-of-basis matrix corresponds to swapping $\mathcal{B}'$ and $\mathcal{B}$.

Proof. If $\mathcal{B} = (v_1, \dots, v_n)$, then $I(v_i) = v_i$ and so $[I(v_i)]_{\mathcal{B}} = [v_i]_{\mathcal{B}}$, which is the i -th standard basis vector. Hence ${}_{\mathcal{B}}[I]_{\mathcal{B}} = I$.

For $\mathcal{B}, \mathcal{B}'$ we have

$${}_{\mathcal{B}}[I]_{\mathcal{B}} = {}_{\mathcal{B}}[I]_{\mathcal{B}'} {}_{\mathcal{B}'}[I]_{\mathcal{B}}, \quad {}_{\mathcal{B}'}[I]_{\mathcal{B}'} = {}_{\mathcal{B}'}[I]_{\mathcal{B}} {}_{\mathcal{B}}[I]_{\mathcal{B}'},$$

so ${}_{\mathcal{B}'}[I]_{\mathcal{B}}$ is invertible and its inverse is ${}_{\mathcal{B}}[I]_{\mathcal{B}'}$. $\square$

Corollary 4.9. Let P be the change-of-basis matrix writing source basis $\mathcal{B}'$ in terms of the target basis $\mathcal{B}$, i.e. $P = {}_{\mathcal{B}}[I]_{\mathcal{B}'}$, and let Q be the change-of-basis matrix writing source basis $\mathcal{C}'$ in terms of the target basis $\mathcal{C}$, i.e. $Q = {}_{\mathcal{C}}[I]_{\mathcal{C}'}$. Then

$$\mathcal{C}'[T]_{\mathcal{B}'} = Q^{-1} \cdot \mathcal{C}[T]_{\mathcal{B}} \cdot P.$$

Example 4.10. Let F^2 have the standard basis (e_1, e_2) , and consider another basis

$$v_1 = (1, 1), \quad v_2 = (-1, 1).$$

Assume $\text{char}(F) \neq 2$, so that (v_1, v_2) is indeed a basis.

The change-of-basis matrix writing the new basis $\mathcal{B}' = (v_1, v_2)$ in terms of the standard basis $\mathcal{E} = (e_1, e_2)$ is

$$P = \varepsilon[I]_{\mathcal{B}'} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

whose columns are $[v_1]_{\mathcal{E}}$ and $[v_2]_{\mathcal{E}}$.

Now let $T : F^2 \rightarrow F^2$ be given by $T(a, b) = (-b, 2a)$, so with respect to the standard basis

$$\varepsilon[T]_{\mathcal{E}} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}.$$

We can compute the matrix of T with respect to the basis $\mathcal{B}'$:

$${}_{\mathcal{B}'}[T]_{\mathcal{B}'} = P^{-1} \varepsilon[T]_{\mathcal{E}} P.$$

Reminder. If $A \in M_{m \times n}(F)$ and we consider

$$T_A : F^n \rightarrow F^m, \quad T_A(v) = Av,$$

then with respect to the standard bases $\mathcal{E}_n$ of F^n and $\mathcal{E}_m$ of F^m , we have

$$\varepsilon_m[T_A]_{\mathcal{E}_n} = A,$$

since $T_A(e_i) = Ae_i$ and the i -th column of the matrix is exactly Ae_i .

Rank of a Linear Transformation

Definition 4.11. Let $T : V \rightarrow W$ be linear. We define:

- The *image* (or range) of T :

$$\text{im}(T) := T(V) = \{T(v) : v \in V\} \subseteq W.$$

- The *kernel* (or nullspace) of T :

$$\ker(T) := \{v \in V : T(v) = 0\} \subseteq V.$$

Proposition 4.12. Both $\text{im}(T)$ and $\ker(T)$ are vector subspaces of W and V respectively.

Proof. For the image: if $w_1 = T(v_1)$ and $w_2 = T(v_2)$, then

$$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2) \in \text{im}(T),$$

and for $\lambda \in F$,

$$\lambda w_1 = \lambda T(v_1) = T(\lambda v_1) \in \text{im}(T).$$

For the kernel: if $v_1, v_2 \in \ker(T)$, then

$$T(v_1 + v_2) = T(v_1) + T(v_2) = 0 + 0 = 0,$$

and if $v \in \ker(T)$, then

$$T(\lambda v) = \lambda T(v) = \lambda \cdot 0 = 0.$$

Thus both are subspaces. □

Observe that $\text{im}(T) = W$ if and only if T is surjective. We have a similar characterisation of injectivity of T :

Lemma 4.13. T is injective if and only if $\ker(T) = 0$.

Proof. If $\ker(T) \neq 0$, there is a vector $v \neq 0$ such that $T(v) = 0$, but then $T(v) = T(0)$, so T is not injective. Conversely, if T is not injective, so $T(v) = T(w)$ for some $v \neq w$, then $T(v - w) = 0$, but $v - w \neq 0$, so $\ker(T) \neq 0$. $\square$

Definition 4.14. Let $T : V \rightarrow W$ be a linear map such that $\text{im } T$ is finite-dimensional. Then we define the *rank* of T to be:

$$\text{rk}(T) := \dim(\text{im}(T)).$$

Example 4.15. • If $T = \lambda I_V : V \rightarrow V$, and V is finite-dimensional, then $\text{rk}(T) = \dim V$ unless $\lambda = 0$, in which case $\text{rk}(T) = 0$.

- If T is surjective and W is finite-dimensional, then $\text{rk}(T) = \dim W$.
- For $T = \frac{d}{dx} : F[x]_{\leq m} \rightarrow F[x]_{\leq m-1}$, the map is surjective, so $\text{rk}(T) = m$.
- If T is given on F^2 by

$$T(a, b) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = (b, 0),$$

then $\text{im}(T)$ is the x -axis, which has dimension 1, so $\text{rk}(T) = 1$.

Remark 4.16. The example of differentiation shows that

$$\frac{d}{dx} : F[x] \rightarrow F[x]$$

is surjective but not injective (constants map to 0). This cannot happen for a linear map $T : V \rightarrow V$ when V is finite-dimensional, as we will see via rank-nullity.

Proposition 4.17. (i) Let $A \in M_{m,n}(F)$ and let $T_A : F^n \rightarrow F^m$ be the linear map $T_A(v) = Av$. Then:

$$\text{rk}(T_A) = \text{rk}(A).$$

- (ii) Let $T : V \rightarrow W$ be a linear map between finite-dimensional vector spaces V, W with bases $\mathcal{B}, \mathcal{C}$, respectively. Then:

$$\text{rk}({}_c[T]_{\mathcal{B}}) = \text{rk}(T).$$

Proof. (i) The image of T_A is

$$\{Av : v \in F^n\} = \text{colspace}(A),$$

so $\dim \text{im}(T_A) = \dim \text{colspace}(A) = \text{rk}(A)$.

(ii) Let V, W be finite-dimensional with bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$. Then the columns of the matrix ${}_c[T]_{\mathcal{B}}$ are exactly the coordinate vectors

$$[T(v_1)]_{\mathcal{C}}, \dots, [T(v_n)]_{\mathcal{C}}.$$

Thus the column space of ${}_c[T]_{\mathcal{B}}$ is

$$\text{Span}([T(v_1)]_{\mathcal{C}}, \dots, [T(v_n)]_{\mathcal{C}}) = [\text{im}(T)]_{\mathcal{C}},$$

the coordinate image of $\text{im}(T)$ under the isomorphism $[\cdot]_{\mathcal{C}} : W \rightarrow F^m$. Since $[\cdot]_{\mathcal{C}}$ is an isomorphism, by Lemma 4.18 below,

$$\dim \text{im}(T) = \dim [\text{im}(T)]_{\mathcal{C}} = \dim \text{colspace}({}_c[T]_{\mathcal{B}}).$$

Hence $\text{rk}(T) = \text{rk}({}_c[T]_{\mathcal{B}})$. □

Lemma 4.18. If $T : V \rightarrow W$ is an isomorphism, then $\dim V = \dim W$.

Proof. Let $(v_1, \dots, v_n)$ be a basis of V . Since T is bijective, the vectors

$$T(v_1), \dots, T(v_n)$$

are linearly independent (injectivity) and span W (surjectivity). Hence they form a basis of W , so $\dim W = n = \dim V$. □

Corollary 4.19. If $T : V \rightarrow W$ is a linear map and $U \subseteq V$ is a subspace such that $T|_U$ is injective, then

$$\dim U = \dim T(U).$$

Proof. Restrict T to U . Since $T|_U$ is injective, $T|_U : U \rightarrow T(U)$ is bijective, hence an isomorphism. By the lemma, $\dim U = \dim T(U)$. □

Theorem 4.20 (Rank–Nullity). Let $T : V \rightarrow W$ be linear, where V is finite-dimensional. Then

$$\dim V = \text{rk}(T) + \dim \ker(T).$$

Proof. We first choose a basis of the kernel. Let

$$(v_1, \dots, v_m)$$

be a basis of $\ker(T)$. Extend this to a basis of V :

$$(u_1, \dots, u_k, v_1, \dots, v_m),$$

so that

$$\dim V = k + m.$$

We claim that $(T(u_1), \dots, T(u_k))$ is a basis of $\text{im}(T)$.

Linear independence: Suppose

$$\lambda_1 T(u_1) + \dots + \lambda_k T(u_k) = 0.$$

Then

$$T(\lambda_1 u_1 + \dots + \lambda_k u_k) = 0,$$

so $\lambda_1 u_1 + \dots + \lambda_k u_k \in \ker(T)$. But the set

$$(u_1, \dots, u_k, v_1, \dots, v_m)$$

is a basis of V , so the only way a linear combination of $u_1, \dots, u_k$ lies in $\text{Span}(v_1, \dots, v_m)$ is for all λ_i to be zero. Hence $T(u_1), \dots, T(u_k)$ are linearly independent.

Spanning: For any $v \in V$ we can write

$$v = \sum_{i=1}^k \alpha_i u_i + \sum_{j=1}^m \beta_j v_j.$$

Applying T gives

$$T(v) = \sum_{i=1}^k \alpha_i T(u_i) + \sum_{j=1}^m \beta_j T(v_j) = \sum_{i=1}^k \alpha_i T(u_i),$$

since $T(v_j) = 0$ for all j . Thus every element of $\text{im}(T)$ is a linear combination of $T(u_1), \dots, T(u_k)$, so they span $\text{im}(T)$.

Hence

$$\text{rk}(T) = \dim \text{im}(T) = k,$$

and therefore

$$\dim V = k + m = \text{rk}(T) + \dim \ker(T).$$

□

Structure given by the proof. The proof shows that we can choose bases

$$\mathcal{B} = (u_1, \dots, u_k, v_1, \dots, v_m) \quad \text{of } V,$$

with $(v_1, \dots, v_m)$ a basis of $\ker T$ and $(u_1, \dots, u_k)$ mapping to a basis $(T(u_1), \dots, T(u_k))$ of $\text{im}(T)$.

If W is also finite-dimensional, we can extend $(T(u_1), \dots, T(u_k))$ to a basis of W :

$$\mathcal{C} = (T(u_1), \dots, T(u_k), w_{k+1}, \dots, w_p).$$

With respect to these bases $\mathcal{B}$ and $\mathcal{C}$ we obtain a particularly simple matrix:

$${}_c[T]_{\mathcal{B}} = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

where I_k is the $k \times k$ identity matrix and $k = \text{rk}(T)$.

Remark 4.21. This shows that a linear map $T : V \rightarrow W$ of rank k can always be written, in suitable bases of V and W , as the block matrix

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}.$$

Conversely, the rank of such a matrix is k , so the rank is the unique integer k such that ${}_c[T]_{\mathcal{B}}$ can be brought to this form.

Corollary 4.22. Let $A \in M_{m \times n}(F)$, and let $T_A : F^n \rightarrow F^m$ be the associated linear map. Then there exist invertible matrices $Q \in GL_m(F)$, $P \in GL_n(F)$ such that

$$Q^{-1}AP = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

where $k = \text{rk}(A)$. In other words, A is equivalent to this canonical form under row and column operations.

Row and column operations viewpoint. Using Gaussian elimination, we can perform row operations to obtain a row echelon form, and then column operations to reach the block matrix

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}.$$

Remark 4.23. Given any matrix $A \in M_{m,n}(F)$ of rank k , if B and C are the standard bases of F^n and F^m , respectively, then ${}_C[T_A]_B = A$. By the above we can find bases C', B' such that ${}_{C'}[T_A]_{B'} = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}$, in block form. We conclude that:

$$A = Q \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix} P^{-1},$$

where $Q = {}_C[I]_{C'}$ and $P = {}_B[I]_{B'}$. In other words, we have:

$$\begin{aligned} &\{\text{Matrices } A \in M_{m,n}(F) \text{ of rank } k\} \\ &= \{Q \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix} P^{-1} : Q \in M_{m,m}(F), P \in M_{n,n}(F) \text{ are invertible}\}. \end{aligned}$$

A similar consequence is that, for any finite-dimensional vector spaces V and W with $\dim V = n$ and $\dim W = m$, and an arbitrary linear transformation $T : V \rightarrow W$, the possible matrices ${}_C[T]_B$ one can get, as B and C run over all bases of V and W , respectively, are precisely

$${}_C[T]_B = Q^{-1} \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix} P,$$

as Q and P run over all invertible matrices of sizes $m \times m$ and $n \times n$, respectively (and they are the change-of-basis that goes from ${}_C[T]_B$ to the matrix $\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}$). By the preceding paragraph, ${}_C[T]_B$ can therefore be any matrix of rank k and size $m \times n$. So the set of possible matrices one can get, allowing B and C to vary, depend only on the dimensions of V and W and the rank of T , and nothing else.

Corollary 4.24. Let $T : V \rightarrow W$ be linear, with V and W finite-dimensional. The following are equivalent:

- (1) T is bijective (an isomorphism).
- (2) T is injective and $\dim V = \dim W$.
- (3) T is surjective and $\dim V = \dim W$.

Proof. (1) $\Rightarrow$ (2),(3): If T is an isomorphism, it is in particular injective and surjective. By the lemma above, $\dim V = \dim W$.

(2) $\Rightarrow$ (1): If T is injective, then $\ker T = \{0\}$, so by rank-nullity

$$\dim V = \text{rk}(T) + \dim \ker T = \text{rk}(T).$$

But $\dim V = \dim W$, so $\text{rk}(T) = \dim W$. Hence $\text{im}(T)$ is a subspace of W of full dimension, so $\text{im}(T) = W$, i.e. T is surjective as well. Thus T is bijective.

(3) $\Rightarrow$ (1): If T is surjective, then $\text{im}(T) = W$, so

$$\text{rk}(T) = \dim W = \dim V.$$

By rank-nullity,

$$\dim V = \text{rk}(T) + \dim \ker T$$

forces $\dim \ker T = 0$, hence $\ker T = \{0\}$ and T is injective. So T is bijective. $\square$

We can also interpret this in terms of left and right inverses:

Proposition 4.25. Given a linear transformation $T : V \rightarrow W$ for V and W finite dimensional:

1. There exists a left inverse $S : W \rightarrow V$, i.e., such that $S \circ T = I_V$, if and only if T is injective;
2. There exists a right inverse $S : W \rightarrow V$, i.e., $T \circ S = I_W$, if and only if T is surjective.

Proof. 1. First, if there exists a left inverse, then $S \circ T = I_V$ is injective, so T must also be injective. Conversely, if T is injective, $\ker T = 0$, and pick a basis $B = (v_1, \dots, v_n)$ of V . We then have that $Tv_1, \dots, Tv_n$ is linearly independent and can be extended to a basis $(Tv_1, \dots, Tv_n, w_1, \dots, w_{m-n})$ of W . We then define $S(Tv_i) = v_i$ and $S(w_i) = 0$, and this uniquely extends to a linear map $S : W \rightarrow V$ with the property that $S(T(v_i)) = v_i$ for all i , and hence $S \circ T = I_V$, as $(v_1, \dots, v_n)$ is a basis.

2. First, if there exists a right inverse, then $T \circ S = I_W$ is surjective, so T must also be surjective. Conversely, if T is surjective, pick a basis $B = (v_1, \dots, v_n, u_1, \dots, u_k)$ of V such that $(u_1, \dots, u_k)$ is a basis for $\ker T$. So $(Tv_1, \dots, Tv_n)$ is a basis of C , as above. We can then define $S(Tv_i) = v_i$ for all i , and this uniquely extends to a linear map $S : W \rightarrow V$ with the property that $T \circ S(Tv_i) = Tv_i$ for all i , so $T \circ S = I_W$ as $(Tv_1, \dots, Tv_n)$ is a basis of W . $\square$

Corollary 4.26. For V, W finite-dimensional vector spaces, and $T : V \rightarrow W$ a linear transformation, the following are equivalent:

1. T is bijective (i.e., an isomorphism);
2. T has a left inverse and $\dim V = \dim W$;
3. T has a right inverse and $\dim V = \dim W$.

Remark 4.27. As before, if T is invertible, any left inverse is the inverse, and any right inverse is the inverse.

Taking matrices, we recover what we knew before, that square matrices have an inverse if and only if they have *either* a left or a right inverse (and, as we saw, every left or right inverse in this case is an inverse).

Dual vector spaces

Definitions and basic properties

Definition 4.28. $V^* := \{T : V \rightarrow F \text{ linear}\}$ is the vector space of linear maps from V to F .

The addition is $(f + g)(v) = f(v) + g(v)$ and the scalar multiplication is $(\lambda f)(v) := \lambda(f(v))$. This is also denoted as $\text{Hom}(V, F)$; more generally $\text{Hom}(V, W)$ denotes the vector space of linear maps from V to W .

Exercise 4.29. Verify the vector space axioms for $\text{Hom}(V, W) := \{T : V \rightarrow W \text{ linear}\}$, and hence also for V^* .

Example 4.30. Consider the case $V = F^n$, viewed as column vectors. Then we have seen that all linear maps $V \rightarrow F \cong F^1$ are given by multiplication by matrices, in this case row vectors $u \in M_{1,n}(F)$. In other words, if we think of V as the column vectors $M_{n,1}(F)$, then we can identify the dual with row vectors $M_{1,n}(F)$, so that if $u \in M_{1,n}(F)$ and $v \in M_{n,1}(F)$, we get the linear transformation $T_u : M_{n,1}(F) \rightarrow M_{1,1}(F) \cong F$. Precisely, this gives an isomorphism $M_{n,1}(F) \rightarrow M_{1,n}(F)^*$, $u \mapsto T_u$ (omitting the identification $M_{1,1}(F) \cong F$, $(\lambda) \mapsto \lambda$).

The preceding example shows that dualising a finite-dimensional vector space produces another which looks very similar, and in particular has the same dimension (we will explain this in more detail later, but this follows because $\dim V = n$ implies $V \cong F^n$, and then $V^* \cong (F^n)^*$).

In addition to dualising vector spaces, we can also dualise maps.

Definition 4.31. Given a linear transformation $T : V \rightarrow W$, the dual transformation $T^* : W^* \rightarrow V^*$ is defined as $T^*(f) := f \circ T$.

Proposition 4.32. Given a linear transformation T , the dual T^* is also a linear transformation.

Proof. We have $T^*(f + g) = (f + g) \circ T = (f \circ T) + (g \circ T) = T^*f + T^*g$, and also $T^*(\lambda f) = T \circ \lambda f = \lambda T \circ f = \lambda T^*(f)$. $\square$

Example 4.33. Let $T = T_A : F^n \rightarrow F^m$. Then, identifying $F^n \cong M_{n,1}(F)$ and $(F^n)^* \cong M_{1,n}(F)$, we have $T_A^*(u)v = u(Av)$ which implies $T_A^*u = uA$, i.e., $T_A^* : M_{1,m}(F) \rightarrow M_{1,n}(F)$ is the operation of right multiplication by A on row vectors.

Proposition 4.34. Given a composition

$$U \xrightarrow{S} V \xrightarrow{T} W,$$

$$U^* \xleftarrow{S^*} V^* \xleftarrow{T^*} W^*,$$

we have $(T \circ S)^* = S^* \circ T^*$. Also, $I_V^* = I_{V^*}$.

Proof. The first assertion is just associativity of composition: $(T \circ S)^*(f) = f \circ (T \circ S) = (f \circ T) \circ S = S^*(T^*(f))$. The second follows from the identity for composition: $f \circ I_V = f$. $\square$

Corollary 4.35. If $T : V \rightarrow W$ is invertible, so is T^* , and $(T^{-1})^* = (T^*)^{-1}$.

Proof. We have $T^* \circ (T^{-1})^* = (T^{-1} \circ T)^* = I_V^* = I_{V^*}$. Similarly $(T^{-1})^* \circ T^* = I_W^* = I_{W^*}$. $\square$

Double dual subspace

Given a vector space V , we define the map

$$\iota : V \rightarrow (V^*)^*, v \mapsto \iota_v, \quad \iota_v(f) = f(v). \quad (4.36)$$

Notice that no basis is required in order to define this map!

Proposition 4.37. The map ι is injective and linear (assuming V is finite-dimensional).²

Proof. The linearity follows because $\iota_{v+v'}(f) = f(v + v') = f(v) + f(v') = \iota_v(f) + \iota_{v'}(f)$, and similarly $\iota_{\lambda v}(f) = \lambda \iota_v(f)$.

For injectivity, suppose that $v \in V$ is nonzero. Then we have an inclusion $F \cdot v = \text{Span}(v) \subseteq V$ of subspaces, so by Proposition A.10, the restriction map $V^* \rightarrow (F \cdot v)^*$ is surjective. That is, there exists $f \in V^*$ with $f(v) \neq 0$. Then $\iota_v(f) = f(v) \neq 0$ as well, so $\iota_v \neq 0$. Thus, we conclude that $\ker(\iota) = 0$, so that ι is injective, as it is linear. $\square$

We will see that ι is actually an isomorphism when V is finite-dimensional. To do this we will find a basis for V^* , which is important on its own and very useful for writing matrices.

Definition 4.38. Given a finite basis $(v_1, \dots, v_n)$ of a vector space V over F , let $v_1^\vee, \dots, v_n^\vee \in V^*$ be the functions $v_i^\vee(\lambda_1 v_1 + \dots + \lambda_n v_n) = \lambda_i$.

In particular, we have the useful formula

$$v_i^\vee(v_j) = \delta_{ij} := \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases} \quad (4.39)$$

Proposition 4.40. The n -tuple $(v_1^\vee, \dots, v_n^\vee)$ of elements of V^* is a basis.

Proof. We first show that the tuple is linearly independent. Indeed, if $f := \lambda_1 v_1^\vee + \dots + \lambda_n v_n^\vee = 0$ then $f(v_i) = \lambda_i = 0$ for all i . Next we show that it spans. If $f \in V^*$, we claim that

$$f = f(v_1)v_1^\vee + \dots + f(v_n)v_n^\vee. \quad (4.41)$$

Let g be the left-hand side. Then $g(v_i) = f(v_i)$ for all i , by definition. By linearity and the fact that $(v_1, \dots, v_n)$ span, we get that $g = f$. $\square$

Definition 4.42. The basis $B^\vee := (v_1^\vee, \dots, v_n^\vee)$ is called the *dual basis* of $B = (v_1, \dots, v_n)$.

It will be useful later to note a companion formula for (4.41): for all $v \in V$,

$$v = v_1^\vee(v)v_1 + \dots + v_n^\vee(v)v_n. \quad (4.43)$$

We immediately deduce:

Corollary 4.44. For V a finite-dimensional vector space, V^* is also finite-dimensional, and $\dim V^* = \dim V$.

This together with Proposition 4.37 yields:

Corollary 4.45. For V a finite-dimensional vector space, $\iota : V \rightarrow (V^*)^*$ is an isomorphism.

Proof. We have seen that ι is linear and injective. Since V is finite-dimensional, so is $(V^*)^*$, and it has the same dimension. Now ι is a linear map between finite-dimensional vector spaces of the same dimension; so it is an isomorphism. $\square$

²Finite-dimensionality is not required for the statement to be true; the proof in general requires one to work with bases for infinite-dimensional vector spaces.

Double dual basis

Given a finite dimensional vector space V and a basis $B \subseteq V$, we can form bases:

$$B^\vee \subseteq V^*, B^{\vee\vee} \subseteq V^{**}.$$

But we also have an isomorphism:

$$\iota : V \xrightarrow{\sim} (V^*)^*$$

Proposition 4.46. Taking double dual basis recovers the image under the isomorphism ι :

$$B^{\vee\vee} = \iota(B).$$

Proof. Let $B = (v_1, \dots, v_n)$, $B^\vee = (v_1^\vee, \dots, v_n^\vee)$, and $B^{\vee\vee} = (v_1^{\vee\vee}, \dots, v_n^{\vee\vee})$. Then we have:

$$\iota(v_i)(v_j^\vee) = v_j^\vee(v_i) = \delta_{ij} = v_i^{\vee\vee}(v_j^\vee).$$

Now fix i . Since $\iota(v_i)(v_j^\vee) = v_i^{\vee\vee}(v_j^\vee)$ for all j , by taking linear combinations of the basis $(v_1^\vee, \dots, v_n^\vee)$ of V^* , we get $\iota(v_i)(f) = v_i^{\vee\vee}(f)$ for all $f \in V^*$. That is, $\iota(v_i) = v_i^{\vee\vee}$. $\square$

As a consequence, $v_i^{\vee\vee}$ depends only on v_i , not on v_j for $j \neq i$! This is *not true* about the dual basis: $v_i^\vee$ depends importantly on all of the v_j . Indeed, it is defined so that $v_i^\vee(v_j) = 0$ when $i \neq j$.

Example 4.47. Let $V = F^2$, with basis $B = (v_1, v_2)$ for $v_1 = e_1$ and $v_2 = \lambda e_1 + e_2$. Then $v_1^\vee = e_1^\vee - \lambda e_2^\vee$. We see that it indeed depends on λ , hence on v_2 .

In terms of column and row vectors: $B = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \lambda \\ 1 \end{pmatrix} \right)$ and $B^\vee = \left(\begin{pmatrix} 1 & 0 \end{pmatrix}, \begin{pmatrix} -\lambda & 1 \end{pmatrix} \right)$.

Orthogonal subspaces

Given $W \subseteq V$, we define $W^\perp \subseteq V^*$ by:

Definition 4.48. $W^\perp := \{f \in V^* \mid f(w) = 0, \forall w \in W\}$.

In other words, it is the set of vectors which *annihilate* all of W (are zero on all of W).

Example 4.49. Suppose that V has basis $B = (v_1, \dots, v_n)$ and $W = \text{Span}(v_1, \dots, v_k)$ for some $1 \leq k \leq n$. Then $W^\perp = \text{Span}(v_{k+1}^\vee, \dots, v_n^\vee)$. For example, if $V = M_{n,1}(F)$ and $v_i = e_i$ is the standard basis, then

$$W = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_m \\ 0 \\ \vdots \\ 0 \end{pmatrix} : a_1, \dots, a_m \in F \right\}, \quad W^\perp = \{ (0 \quad \dots \quad 0 \quad b_1 \quad \dots \quad b_{n-m}) : b_1, \dots, b_{n-m} \in F \}$$

Proposition 4.50. If V is finite-dimensional, and $W \subseteq V$ a subspace, then

$$\dim V = \dim W + \dim W^\perp.$$

Remark 4.51. When you learn about dot products / inner products, you see that can give an isomorphism $V \rightarrow V^*$, and using this, the preimage of $W^\perp$ will just be the usual subspace of vectors which dot with all of $w \in W$ to zero.

Dualisation and transpose

Theorem 4.52. Let V and W be finite-dimensional vector spaces with bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Let $T : V \rightarrow W$ be a linear transformation. Then

$${}_{\mathcal{B}^\vee}[T^*]_{\mathcal{C}^\vee} = {}_{\mathcal{C}}[T]_{\mathcal{B}}^t. \quad (4.53)$$

Proof. Let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$, with dual bases $\mathcal{B}^\vee = (v_1^\vee, \dots, v_n^\vee)$ and $\mathcal{C}^\vee = (w_1^\vee, \dots, w_m^\vee)$. Then

$$T^*(w_j^\vee)(v_i) = w_j^\vee \circ T(v_i) = w_j^\vee(T(v_i)).$$

By definition and (4.41), the left-hand side is the ij -th entry of the matrix ${}_{\mathcal{B}^\vee}[T^*]_{\mathcal{C}^\vee}$, whereas by (4.43), the right-hand side is the ji -th entry of the matrix ${}_{\mathcal{C}}[T]_{\mathcal{B}}$. $\square$

Example 4.54. In the case $V = F^n, W = F^m$, we can also verify the theorem directly from Example 4.33: Every linear transformation from V to W is of the form T_A for $A \in M_{m,n}(F)$. We already knew the formula ${}_{(e_1, \dots, e_m)}[T_A]_{(e_1, \dots, e_n)} = A$. If we identify F^n with column vectors, and $(F^n)^*$ with row vectors, then for $u \in M_{1,n}(F)$ a row vector, thought of as an element of V^* , we have

$$[u]_{(e_1^\vee, \dots, e_n^\vee)} = u^t.$$

We found in Example 4.33 that $T_A^*(T_u) = T_{uA}$, left multiplication by the row vector $uA \in M_{1n}(F) \cong V^*$. We then get:

$$[T_A^*u]_{(e_1^\vee, \dots, e_n^\vee)} = [uA]_{(e_1^\vee, \dots, e_n^\vee)} = (uA)^t = A^t u^t = A^t [u]_{(e_1^\vee, \dots, e_n^\vee)}.$$

This implies (4.53), by the definition of the matrix of a linear transformation.

A Bonus material (not to cover in lecture): quotient vector spaces

Note: This topic will not be assumed knowledge for either the January exam or the final exam.

Let V be a vector space over a field F and let $U \subseteq V$ be a subspace. We define an equivalence relation on V by $v \sim v'$ if $v' - v \in U$. The equivalence classes are the sets $v + U := \{v + u : u \in U\} \subseteq V$ for each $v \in V$; these are also called cosets of U in V . Note that $v + U = v' + U$ if and only if $v' - v \in U$.

Definition A.1. The quotient $V/U := \{v + U : v \in V\}$ is the set of equivalence classes under the equivalence relation above.

Lemma A.2. The quotient V/U is naturally a vector space, under addition $(v + U) + (v' + U) = (v + v') + U$ and scalar multiplication $\lambda(v + U) = (\lambda v) + U$.

Proof. We need to check the axioms. The zero element of V/U is U : $U + (v + U) = v + U$ for all v . The axioms of a commutative group follow from those of V : commutativity, associativity, and the negation are all the same: $(v + U) + (v' + U) = (v + v') + U = (v' + v) + U = (v' + U) + (v + U)$, and similarly $((v + U) + (v' + U)) + (v'' + U) = (v + v' + v'') + U = (v + U) + ((v' + U) + (v'' + U))$, and $((v + U) + (-v + U)) = U$. Similarly, the associativity of scalar multiplication, the unit axiom, and the distributivity properties all follow from those of V . $\square$

Theorem A.3 (The First Isomorphism Theorem, for vector spaces). Given a linear transformation $T : V \rightarrow W$, we have an isomorphism $\bar{T} : V/\ker T \rightarrow \text{im } T$, given by $\bar{T}(v + \ker T) = T(v)$.

Proof. We first check that $\bar{T}$ is well-defined: if $v + \ker T = v' + \ker T$, i.e., $v - v' \in \ker T$, then $\bar{T}(v' + \ker T) = T(v') = T(v) = \bar{T}(v + \ker T)$.

Next we check that $\bar{T}$ is linear: $\bar{T}((v + \ker T) + (v' + \ker T)) = T(v) + T(v') = T(v + v') = \bar{T}((v + v') + \ker T)$. Similarly, $\bar{T}(\lambda(v + \ker T)) = \bar{T}(\lambda v + \ker T) = T(\lambda v) = \lambda T(v) = \lambda \bar{T}(v + \ker T)$.

We check that $\bar{T}$ is injective: it is equivalent to show that its kernel is zero, since it is linear. Now $\bar{T}(v + \ker T) = 0$ if and only if $T(v) = 0$, i.e., $v \in \ker T$, so that $v + \ker T = \ker T$.

Finally we check surjectivity. If $w \in \text{im } T$ then $w = Tv$ for some v , and hence $w = \bar{T}(v + \ker T)$. $\square$

As a consequence, $\text{rk } T = \dim \text{im } T = \dim V/\ker T$. In other words, the rank-nullity theorem (which we proved only for finite-dimensional vector spaces) can be rewritten, using the first isomorphism theorem, as the following (which we only have proved for V finite-dimensional):

$$\dim V = \dim(V/\ker T) + \dim \ker T. \quad (\text{A.4})$$

This statement is more intuitive. We can generalise it to:

Proposition A.5. Let V be a vector space and $U \subseteq V$ a subspace. Then $\dim V = \dim U + \dim V/U$.

Proof. We can use the projection map $\pi : V \rightarrow V/U$, $\pi(v) = v + U$. This map is clearly surjective and linear, and the kernel is U since $v + U = U$ if and only if $v \in U$. Thus the result follows from (A.4). $\square$

Remark A.6. Alternatively, we could prove this directly (rehashing the proof of the rank-nullity theorem): take a basis $(u_1, \dots, u_m)$ of U , extend to a basis $(u_1, \dots, u_m, v_1, \dots, v_n)$ of V , and then observe that $v_1 + U, \dots, v_n + U$ form a basis for V/U .

Remark A.7. Note that in the case that $F = \mathbb{F}_p$ then $|F^n| = p^n$. So if $\dim V = n$, $V \cong F^n$, and hence $|V| = p^n$. Thus, for $F = \mathbb{F}_p$ and V finite-dimensional, (A.4) becomes equivalent to the statement

$$|V| = |V/U| \cdot |U|.$$

This suggestive equality will also hold in the more general context of subgroups of finite groups, as you will learn in Term 2.

Observe that every linear map $T : V \rightarrow W$ factors as a composition of linear maps as follows:

$$V \xrightarrow{\pi} V/\ker T \xrightarrow{\overline{T}} \operatorname{im} T \hookrightarrow W, \quad (\text{A.8})$$

and so everything can be reduced to the three cases:

- projections $\pi : V \rightarrow V/U$, the content of the rank-nullity theorem, where it becomes Proposition A.5;
- isomorphisms, where the rank-nullity theorem reduces to saying that the dimension of the source and target are the same,
- inclusions $i : U \hookrightarrow V$, where the rank-nullity theorem is trivial.

Dual nature of subspaces and quotient vector spaces, and the orthogonal subspace

Recall that, given a subspace U of a vector space V , the orthogonal subspace of V^* is defined as $U^\perp := \{f \in V^* \mid f(u) = 0, \forall u \in U\} \subseteq V^*$.

Proposition A.9. There is an isomorphism $(V/U)^* \rightarrow U^\perp$ given by the map $\pi^* : (V/U)^* \rightarrow V^*, f \mapsto f \circ \pi$.

Proof. First note that $f \circ \pi(u) = 0$ for all $u \in U$. Thus the map $(V/U)^* \rightarrow U^\perp$ is well defined, and by definition it is linear.

It is injective because, if $f \in (V/U)^* \neq 0$, then there must be some $v \in V$, with such that $f(v + U) \neq 0$. But then $f \circ \pi(v) \neq 0$ as well, hence $\pi^*(f) \neq 0$.

Finally, we check surjectivity. If $g \in U^\perp$, we can consider $f \in (V/U)^*$ with $f(v + U) := g(v)$. This is well-defined because $g(u) = 0$ for all $u \in U$, and $g = \pi^*(f)$. $\square$

Thus, the dual vector space to a quotient vector space is naturally a subspace of the dual vector space, just the orthogonal. The other way is slightly more work: Consider the inclusion map $i : U \rightarrow V$ for a subspace U of a vector space V . Then the dual $i^* : V^* \rightarrow U^*$ is just the restriction map: $i^*(f) = f \circ i$ just sends $u \in U$ to $f(u)$, i.e, $f|_U$.

Proposition A.10. The map i^* is surjective, with kernel $U^\perp$ (assuming V is finite-dimensional).³

By the first isomorphism theorem, this has the immediate consequence:

Corollary A.11. We have an isomorphism $V^*/U^\perp \rightarrow U^*, f + U^\perp \mapsto f|_U$.

³Finite-dimensionality is not required for the statement to be true; the proof in general requires one to work with bases for infinite-dimensional vector spaces.

Proof of Proposition A.10. Given $f \in U^*$, we can form a basis for U , $(u_1, \dots, u_m)$ (Since V is finite-dimensional and $U \subseteq V$, U is also finite-dimensional), and extend it to a basis $(u_1, \dots, u_m, v_1, \dots, v_n)$ for V . Then we can extend f to the linear map $\tilde{f}(\lambda_1 u_1 + \dots + \lambda_m u_m + \nu v_1 + \dots + \nu_n v_n) = f(\lambda_1 u_1 + \dots + \lambda_m u_m)$.

The statement about the kernel follows by definition of $U^\perp$ ($f|_U \neq 0$ if and only if $f \in U^\perp$). $\square$

As a result, the dual of a subspace identifies with a quotient of the dual space.